

REQUEST FOR PROPOSAL (RFP) RESPONSE

State Integrated Mobility Charging Platform (SIMCP)

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Table of Contents

Document Control	1
Table of Contents	3
List of Figures.....	6
List of Tables	6
Executive Summary	7
1. Background & Problem Statement.....	7
1.1 Current State.....	7
1.2 Business Problem	7
2. Vision, Objectives & Success Criteria	8
2.1 Vision.....	8
2.2 Strategic Policy Objectives	8
2.3 Success Criteria	10
Phase 0 – Program Foundation	10
Phase 1 – Core Services Foundation Build	10
Phase 2 – Core Modes & Billing (Greater Sydney minimum viable product (MVP))	12
Phase 3 – Expansion & Optimisation (NSW-wide)	14
3. Solution Overview	16
3.1 SIMCP Concept.....	16
3.2 High-Level Architecture Diagram	17
4. Functional Solution Design and Implementation Technology	17
Module 1 – Digital Identity & Mobility Account Platform	17
Purpose.....	17
Core Functions.....	18
Inputs.....	22
Outputs.....	23
Key Business Rules / Logic	23
Interactions with Other Modules	24
Implementation Technology (Indicative)	25
Governance & Compliance	26
Summary.....	26
Module 2 – Mobility Event Capture & Sensing Engine	27
Module Overview	27
Core Functions.....	27
Inputs.....	28
Outputs.....	28

Key Business Rules / Logic	30
Interactions with Other Modules	30
Implementation Technology (Indicative)	31
Governance & Compliance	32
Summary.....	33
Module 3 — Dynamic Pricing & Policy Engine	33
Purpose.....	33
Core Functions.....	34
Inputs.....	35
Outputs.....	35
Key Business Rules / Logic	36
Interactions with Other Modules	36
Implementation Technology (Indicative)	37
Governance & Compliance	39
Summary.....	39
Module 4 — Billing, Aggregation & Settlement System.....	40
Purpose.....	40
Core Functions.....	40
Inputs.....	41
Outputs.....	42
Key Business Rules / Logic	42
Interactions with Other Modules	43
Implementation Technology (Indicative)	43
Governance & Compliance	44
Summary.....	44
Module 5 — Analytics, Optimisation & Governance Platform	44
Purpose.....	44
Core Functions.....	45
Inputs.....	46
Outputs.....	46
Key Business Rules / Logic	46
Phased capability.....	47
Interactions with Other Modules	47
Implementation Technology (Indicative)	47
Governance & Compliance	49
Summary.....	49

5.	System Integration & Interoperability.....	49
6.	Non-Functional Requirements	51
6.1	Performance & Availability	51
6.2	Security & Privacy.....	51
6.3	Usability.....	51
6.4	Maintainability & Operability.....	51
6.5	Future Readiness.....	52
8.	Ethical Considerations	52
9.	Implementation Approach & Roadmap	53
9.1	Delivery Phases	53
9.2	High-Level Project Plan	54
9.2.1	Milestones	54
9.2.2	Dependencies	56
9.2.3	Gantt Chart	57
10.	Risk Analysis & Mitigation	57
11.	Assumptions & Constraints	57
11.1	Legislative & Privacy Assumptions	57
11.2	Scope Boundaries	58
11.3	Legacy System Constraints.....	58
11.4	Rollout & Timeline Constraints.....	58
11.5	Cloud Platform Assumptions	58
11.6	Concession Policy Stability	58
12.	Expected Benefits.....	59
12.1	Benefits to Government.....	59
12.2	Benefits to Residents	59
12.3	Broader Societal Benefits.....	59
13.	Conclusion	60
	Appendices	61
	Appendix 1.....	61
	Appendix 2. Terminology and Definitions	62
	A. Canonical phase definitions (baseline: Section 9.1)	62
	B. Team terminology (quick glossary)	62
	Appendix 3. SIMCP overall workflow (logical data path)	64
	Appendix 4. Module 2 Appendix	64
	A. Event Detection and Validation Fall-back Plan and Architecture	64
	B. Confidence Scoring Mechanism	66

C. Phase 3 Event Detection and Sensing Engine - Implementation of Geo-fencing and Route Matching:	67
Appendix 5. Section 10 Appendix.....	69

List of Figures

Figure 1. High-level architecture diagram of SIMCP system	17
Figure 2. Module 1 - ERD of SIMCP	20
Figure 3. Module 3 - Solution Overview.....	27
Figure 4. Module 3 - System Process of Dynamic Pricing.....	34
Figure 5. Module 4 - Billing, Statement and Settlement Flow	40
Figure 6. Module 5 - High level enterprise architecture diagram for SIMCP	45
Figure 7. Module 5 – AWS Cloud Solution for SIMCP.....	48
Figure 8. System Integration and Interoperability Architecture Diagram for SIMCP.....	49

List of Tables

Table 1. Phase 0 - Success Criteria	10
Table 2. Phase 1 - Success Criteria	11
Table 3. Phase 2 - Success Criteria	13
Table 4. Phase 3 – Success Criteria.....	14
Table 5. ERD of Mobility Account Platform.....	18
Table 6. RBAC and Description	21
Table 7. Concession Types and Description	21
Table 8. Managed Identity Lifecycle and Description	22
Table 9. Module 1 - Solution Inputs	22
Table 10. Module 1 - Solution Outputs	23
Table 11. Module 2 - Solution Core Functions	27
Table 12. Module 2 - Solution Inputs	28
Table 13. Module 2 - Solution Outputs	29
Table 14. Module 2 - Key Business Rules and Logics.....	30
Table 15. Module 3 - Solution Core Functions	34
Table 16. Module 3 – Solution Inputs	35
Table 17. Module 3 – Government-Facing Outputs.....	35
Table 18. Module 3 - User-Facing Outputs.....	36
Table 19. Module 4 - Core Billing Functions.....	41
Table 20. Module 4 - Billing Solution Inputs	41
Table 21. Module 4 - Billing Solution Output	42
Table 22. Module 4 - SIMCP Billing Account Structure	42
Table 23. Module 5 - Solution Core Functions	45
Table 24. Module 5 - Solution Inputs	46
Table 25. Milestones of SIMCP System	54
Table 26. Gantt Chart for SIMCP Project Timeline	57
Table 27. SMICP Risk Analysis and Mitigation.....	69

Executive Summary

1. Background & Problem Statement

1.1 Current State

The transport ecosystem in New South Wales operates as a collection of largely independent systems, each managed by different agencies and service providers. Users are required to interact with multiple platforms, accounts, and processes depending on the transport mode they use.

Key stakeholders include:

- **Service NSW.** Vehicle registration and driver licensing
- **Transport for NSW (Opal).** Public transport ticketing and management, toll payment gateway
- **Transurban and other private operators.** Toll road infrastructure
- **Local councils and operators.** Parking systems and meters
- **Uber, DiDi, Lime.** Rideshare and micromobility services
- **Freight and heavy vehicle schemes.** Road usage and compliance systems

Each system currently functions with distinct user journeys, data structures, and operational logic:

- **Ticketing and payment.** Users interact through mode-specific mechanisms (e.g., smartcards, e-tags, licence plate recognition). Fare calculation is governed by individual system rules, with transaction data stored in isolated operator databases.
- **Vehicle and licensing services.** Managed centrally by Service NSW, but with no real-time integration into actual mobility usage or behaviour.
- **External mobility providers.** Parking, rideshare, and micromobility services operate independently, with minimal interoperability or data exchange across platforms.

Overall, the current ecosystem is fragmented and operator-centric, optimised for individual service delivery rather than seamless, end-to-end user journeys. This fragmentation is not incidental but structural, embedded in how systems are designed and governed.

1.2 Business Problem

Despite the presence of multiple mobility systems across the state, there is **no unified framework** to integrate mobility usage, pricing, and policy enforcement across transport modes. In its current form, the transport ecosystem lacks the integration required to support a cohesive, data-driven mobility strategy.

This core issue manifests across three key dimensions:

- **User Experience.** Users must navigate multiple systems with inconsistent interfaces, payment methods, and account structures, leading to friction and reduced accessibility across transport modes.
- **Operational Efficiency.** Data remains siloed across agencies and providers, constraining visibility and limiting the ability to implement coordinated, cross-platform initiatives or enforce system-wide rules.
- **Strategic Policy Capability.** The absence of integrated, real-time mobility data restricts the government's ability to influence travel behaviour. This limits the effectiveness of initiatives such as congestion pricing, demand management, and incentive-based policy interventions.

2. Vision, Objectives & Success Criteria

2.1 Vision

Our vision is to ***“Deliver a unified, intelligent, and resident-centric mobility ecosystem”***, where all transport interactions are seamlessly connected through a single digital account, enabling frictionless travel, transparent billing, and real-time, policy-driven optimisation.

Under this vision, mobility in NSW evolves from fragmented, operator-centric systems into an integrated, account-based platform, where:

- Users no longer need to manage multiple transport systems
- Journeys are automatically detected, classified, and priced
- All charges are consolidated into a single, transparent view
- Government policies can be dynamically applied across all modes, delivering improved user experience and more efficient, fair pricing outcomes

This vision supports the long-term transition toward:

- Frictionless mobility experiences with fully consolidated billing
- Data-driven transport planning and governance
- Scalable integration of future mobility services (e.g., micro-mobility and autonomous transport)

2.2 Strategic Policy Objectives

The State Integrated Mobility Charging Platform (SIMCP) solution is designed not only as a technology platform, but as a policy enablement framework that supports the State Transport Authority’s objectives of reducing traffic congestion and empower public transportation system.

Congestion Management

SIMCP enables real-time congestion management through dynamic pricing and policy interventions. By leveraging predictive mobility data, the platform allows the government to influence travel behaviour, reduce peak demand, and improve overall network efficiency.

Mode Shift & Network Optimisation

The SIMCP platform enables mode shift and network optimisation through its integrated, account-based architecture. By combining real-time mobility data (Module 2) with dynamic pricing and policy controls (Module 3), the system can deliver targeted incentives and pricing signals across transport modes.

This allows the platform to guide users toward more efficient travel choices and increase adoption of public transport, while redistributing demand across time periods and routes.

Equity & Accessibility

SIMCP ensures equitable access by preserving existing Opal concession categories (e.g., adult, student, senior, and concession groups) while extending coverage across both public and private mobility services.

By embedding consistent concessions, pricing caps, and policy safeguards within a unified account framework, the platform supports an all-resident equity model and ensures fair, inclusive access to mobility across all modes.

Administrative Simplification

By consolidating multiple transport systems into a single platform, SIMCP reduces operational complexity for both users and government agencies, streamlines billing, and improves interoperability across services.

Environmental Sustainability

The platform supports sustainability goals by incentivising low-emission transport options and enabling data-driven environmental policies. Improved visibility of mobility patterns allows for more effective planning and emissions reduction strategies.

2.3 Success Criteria

A total of 30 success criteria has been defined across the full delivery lifecycle of the SIMCP project, spanning Phase 0 through to Phase 3. Each criterion specifies a measurable target, a verification method, a responsible party (Verified by), and a sign-off authority (Approved by), ensuring clear accountability at every stage of delivery.

Phase 0 – Program Foundation

This phase focus on governance and compliance prerequisites that must be satisfied before any technical build commences.

Table 1. Phase 0 - Success Criteria

#	Module	Category	Success criterion	Measurable target	Verification method	Verify PIC	Approval PIC
1	All	Governance	Scope & governance model agreed. All agencies sign off on platform scope, ownership, data-sharing principles, and policy guardrails	Signed governance charter by all key stakeholders	Document sign-off review	STA Program Director	STA Executive Sponsor
2	All	Governance	Target operating model (TOM) defined. Roles for STA, operators, councils, and support/dispute routing confirmed	TOM document approved with zero unresolved ownership gaps	Stakeholder review workshop	SIMCP Integration Program Manager	STA Executive Sponsor
3	All	Compliance	Privacy impact assessment (PIA) completed. PIA covering all data sharing and location tracking use cases approved by privacy authority	PIA approved, zero unresolved high-risk findings	Privacy authority sign-off	STA Legal & Privacy Team	NSW Privacy Commissioner

Phase 1 – Core Services Foundation Build

Phase 1 criteria validate that the core capabilities of all five modules function correctly in a controlled environment, covering identity management, event ingestion, pricing rule configuration, billing aggregation, and baseline operational dashboards.

Table 2. Phase 1 - Success Criteria

#	Module	Category	Success criterion	Measurable target	Verification method	Verify PIC	Approval PIC
4	M1	Identity	Mobility ID creation & Service NSW integration. Users can register and receive a Mobility ID linked to their Service NSW account	100% of test accounts verified; onboarding in < 3 minutes (benchmark: The system process usually takes in seconds, and the ID verification completed in minutes)	User acceptance testing (UAT) with synthetic test users	Module 1 Lead & Quality Assurance (QA) Team	Service NSW Integration Owner
5	M1	Identity	Concession eligibility correctly assigned. All Opal concession categories (senior, student, pension, child) applied accurately at registration	≥ 99% accuracy across all concession categories in test dataset	Regression test with known eligibility cases	Module 1 Lead & QA Team	STA Policy Board
6	M1	Security	Authentication & access control enforced. Multi-factor authenticator (MFA) works, Role-based access control (RBAC) prevents unauthorised access across all role types	Zero privilege escalation in pen test; MFA pass rate = 100%	Penetration test + RBAC matrix validation	Independent Security Auditor	STA CISO
7	M1	Identity	Identity lifecycle transitions function correctly. Registration → Verification → Active → Suspension → Deactivation all work with audit log entries	All lifecycle states tested; 100% audit log completeness	Integration test + audit log review	Module 1 Lead & QA Team	STA Program Director
8	M2	Event capture	Core event ingestion pipeline operational. Opal tap-on/off, toll gantries, and parking events ingested into unified event schema	< 5% event loss rate; all events include required schema fields (benchmark: production-grade streaming pipelines typically achieve well under 1% event loss, making <5% a conservative but acceptable.)	Pipeline integration test with real feeds	Module 2 Lead & QA Team	SIMCP Integration Program Manager
9	M2	Performance	Real-time event processing latency. Mobility events processed and published to downstream modules within Service level agreement (SLA)	P95 end-to-end latency < 2 seconds under normal load. (Benchmark: real-time streaming SLAs typically range from sub-	Load test with simulated peak volume	Module 2 Lead & QA Team	STA Program Director

				second to 5 seconds end-to-end.)			
10	M2	Data quality	Duplicate event detection working. System correctly removes duplicate signals within the defined deduplication window	< 0.1% duplicate events in validated output	Injection test with known duplicate signals	Module 2 Lead & QA Team	SIMCP Integration Program Manager
11	M3	Pricing rules	Pricing rules baseline configured & versioned. Time-of-day, zone, and concession rules correctly applied to test events with full audit trail	100% of rules produce expected output in test scenarios; all changes version-tracked	Rules regression test suite	Module 3 Lead & QA Team	STA Policy Board Secretariat
12	M3	Governance	Human approval workflow for rule changes. No pricing rule reaches production without passing the defined approval and rollback workflow	Zero unauthorised production rule deployments in testing period	Change control audit review	SIMCP Platform Owner	STA Policy Board Secretariat
13	M4	Billing	Monthly statement prototype generated correctly. All chargeable events within a billing cycle aggregated into a single itemised statement per account	Statement totals match source event totals; zero unattributed charges in test run	End-to-end billing reconciliation test	Module 4 Lead & Finance Ops	STA Program Director
14	M4	Billing	Individual & organisation billing correctly separated. Personal vs fleet/business charges allocated correctly under mixed-use scenarios	100% correct allocation in all mixed-use test cases	Test case matrix with known scenarios	Module 4 Lead & Finance Ops	Finance Operations Lead
15	M5	Analytics	Core financial & operational dashboards live. Baseline KPI dashboards showing event volumes, billing outcomes, and error rates available to STA	All Phase 1 KPIs visible in dashboard; data refresh lag < 1 hour	Dashboard UAT with STA stakeholders	Module 5 Lead & QA Team	STA Program Director

Phase 2 — Core Modes & Billing (Greater Sydney minimum viable product (MVP))

Phase 2 criteria target the Greater Sydney MVP, addressing more advanced capabilities including geofencing and route matching proof-of-concept accuracy, end-to-end payment processing, real-time congestion pricing triggers, and the analytics feedback loop to the pricing engine.

Table 3. Phase 2 - Success Criteria

#	Module	Category	Success criterion	Measurable target	Verification method	Verify PIC	Approval PIC
16	M1	Identity	Family group & business account management. Family owners can invite members; fleet managers can manage drivers and vehicles via their account	All role-based scenarios complete without errors in UAT	UAT with representative user groups	Module 1 Lead & QA Team	STA Customer Experience Lead
17	M2	Event capture	Confidence scoring validated across modes. Confidence scores correctly reflect signal reliability (tap > GPS > geofence) and flag low-confidence events	Score distribution matches expected hierarchy in ≥ 95% of test cases	Controlled test journeys across modes	Module 2 Lead & QA Team	SIMCP Platform Owner
18	M2	Event capture	Geofencing PoC: Station entry detection. Geofence triggers correctly detect entry into station/stop areas and generate trip-start events	≥ 90% detection accuracy in pilot zones; false positive rate < 5%	Controlled field test in selected stations	Module 2 Lead & Field Test Team	STA Legal–Privacy + SIMCP Platform Owner
19	M2	Event capture	Route matching PoC: Trip attribution accuracy. Route matching correctly identifies the transport line/service used for a sampled set of known journeys	≥ 85% correct attribution on known test journeys	Blind validation against tap-verified ground truth	Module 2 Lead & Independent Bench marker	SIMCP Platform Owner
20	M3	Pricing	Real-time congestion pricing triggers correctly. Pricing adjustments fire when congestion thresholds are breached, within policy guardrails	100% of trigger conditions produce expected pricing output; no guardrail breaches in test	Simulated congestion event injection test	Module 3 Lead & QA Team	STA Policy Board Secretariat
21	M3	Equity	Concession & equity protections upheld under dynamic pricing. Price caps, concession discounts, and equity safeguards never overridden by dynamic pricing logic	Zero concession or cap violations across all dynamic pricing test scenarios	Adversarial test with edge-case pricing inputs	Module 3 Lead & Independent Auditor	STA Policy Board Secretariat
22	M4	Billing	End-to-end payment option live (Option A or B). Per-transaction or monthly bill pathway functioning end-to-end including dispute handling	100% of pilot transactions successfully billed and traceable; dispute resolution tested	Pilot with controlled user cohort in Greater Sydney	Module 4 Lead & Finance Ops	Finance Operations Lead

23	M4	Billing	Operator settlement outputs generated correctly. Revenue allocation to public transport, toll operators, and councils matches agreed revenue-sharing rules	Settlement totals reconcile to < 0.01% variance vs source events	Reconciliation audit with sample data	Module 4 Lead & Finance Ops	Finance Operations Lead + Participating Operators
24	M5	Analytics	Near-real-time operational dashboards live. Live KPI dashboards covering all MVP modes available for STA operations team	Dashboard latency < 5 minutes; all MVP KPIs visible and validated against source data	STA operations team acceptance test	Module 5 Lead & QA Team	STA Program Director
25	M5	Analytics	Pricing effectiveness feedback loop to M3 established. M5 produces versioned pricing effectiveness packs comparing pre/post policy change outcomes	First effectiveness pack delivered within 4 weeks of pilot go-live; approved by policy team	Policy team review & sign-off	Module 5 Lead	STA Policy Board Secretariat

Phase 3 – Expansion & Optimisation (NSW-wide)

Phase 3 criteria set the bar for statewide production readiness, including geofencing and route matching at scale, measurable mode-shift outcomes, onboarding of shared mobility providers, and an independent equity audit confirming no adverse impact on concession or low-income cohorts.

Table 4. Phase 3 – Success Criteria

#	Module	Category	Success criterion	Measurable target	Verification method	Verify PIC	Approval PIC
26	M2	Event capture	Geofencing production accuracy: statewide. Geofencing deployed statewide with fallback to tap; accuracy meets production threshold	≥ 95% station detection accuracy; false positive rate < 2%; tap fallback always available	Statewide field validation + ongoing monitoring	Module 2 Lead & Field Test Team	SIMCP Platform Owner + STA CISO
27	M2	Event capture	Route matching production: trip attribution at scale. Route matching supports accurate fare calculation without requiring tap, including error handling	≥ 92% attribution accuracy; all low-confidence events handled via fallback/dispute path. (Benchmark: GPS-based map matching algorithms typically achieve 83–95% accuracy in practice)	Blind validation + dispute rate monitoring	Module 2 Lead & Independent Bench marker	SIMCP Platform Owner

28	M3	Pricing	Mode-shift measurably achieved via pricing signals. Pricing incentives demonstrably shift demand away from congested modes or time periods	≥ 5% demand shift on targeted congested corridors within 3 months of activation	M5 mode-shift analysis vs baseline	Module 5 Lead (analytics evidence)	STA Policy Board Secretariat
29	M4	Billing	Shared mobility providers onboarded to billing. Bike-share and ride-share events billed correctly within the unified monthly statement	≥ 1 partner live; billing reconciliation < 0.05% variance	Partner onboarding acceptance test	Module 4 Lead & Partner Rep	Finance Operations Lead + SIMCP Integration Program Manager
30	M5	Analytics	Equity analysis confirms no adverse distributional impact. Analytics confirm dynamic pricing does not disproportionately burden concession or low-income cohorts	No statistically significant increase in cost burden for protected cohorts vs baseline	Independent equity audit using M5 outputs	Independent Equity Auditor	STA Executive Sponsor + NSW Privacy Commissioner

3. Solution Overview

3.1 SIMCP Concept

The SIMCP is a unified digital infrastructure that connects every aspect of a resident's mobility life, from catching a bus to paying a toll, into a single, intelligent account. Rather than managing separate apps, cards, and payment methods for each transport service, residents and visitors interact with one platform that automatically handles everything in the background.

At its core, SIMCP operates as a continuous five-step cycle across all transport modes:

Module 1

A resident or visitor first registers through Service NSW, establishing a verified Mobility ID that becomes their single digital identity across the entire transport ecosystem. This account holds their personal details, linked vehicles, concession eligibility, and payment preferences, and it follows them across every journey they make.

Module 2

When travelling, whether tapping on a train, driving through a toll gantry, or parking in a council car park, the platform automatically detects and records the mobility event in real time. Using a combination of existing infrastructure signals (such as Opal readers and toll gantries) and, in later phases, device-based sensing such as GPS and geofencing, every trip is captured, validated, and linked back to the account holder's Mobility ID.

Module 3

Once a mobility event is captured, the platform instantly calculates the appropriate charge based on a rich set of policy rules, including the time of day, the corridor travelled, current congestion levels, and the account holder's concession status. If roads are congested, tolls can be adjusted upward to discourage peak usage; if trains are overcrowded, bus fares can be discounted to encourage mode shift. All pricing decisions are transparent, auditable, and operate within strict government-approved guardrails.

Module 4

At the end of each billing cycle, mobility charges across participating transport modes are consolidated into a single monthly statement, giving account holders one clear view of their mobility spending. SIMCP supports two payment options: Option A, where users continue to pay each operator per transaction while receiving a unified monthly statement, and Option B, our proposed integrated model, where eligible charges are consolidated into one monthly bill through the Service NSW payment gateway. Whether the account holder used public transport, paid a toll, or parked their car, all charge records appear in one place with itemised details for transparency. For business accounts and family groups, charges are automatically allocated to the correct account, member, employee, or vehicle. Revenue allocation outputs are then prepared for the relevant operators and agencies through a structured settlement process.

Module 5

Finally, all these activities feed into a system-wide intelligence layer that gives government planners real-time visibility into how the network is performing. Usage patterns, pricing effectiveness, equity impacts, and congestion trends are continuously monitored, enabling evidence-based policy decisions and ongoing refinement of pricing strategies, closing the loop between operation and policy.

Together, these five modules transform NSW mobility from a fragmented collection of independent systems into a seamless, policy-driven platform – one where residents, visitors, and businesses experience less friction, government gains greater control, and the transport network becomes smarter over time.

3.2 High-Level Architecture Diagram

Figure 1. High-level architecture diagram of SIMCP system

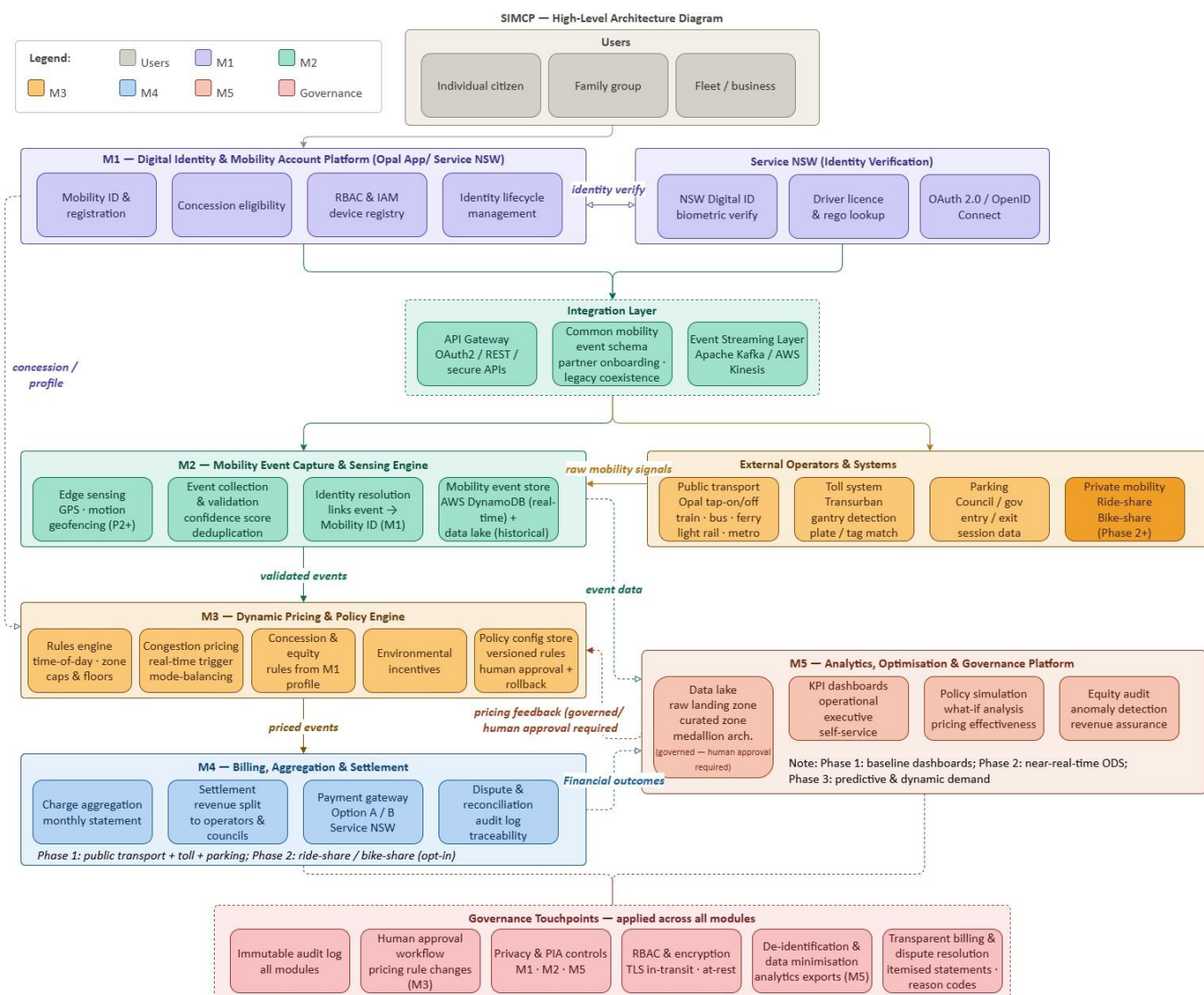


Diagram Source: [SIMCP Architecture diagram.drawio](#)

4. Functional Solution Design and Implementation Technology

Module 1 — Digital Identity & Mobility Account Platform

Purpose

Nowadays, residents in NSW use transport services across multiple platforms such as Opal, toll systems, and parking applications, making it difficult to manage and organise their mobility activities. The purpose of this module is to establish

a unified digital identity and mobility account for each resident, vehicle, and authorised device, addressing the current fragmentation of mobility services across multiple platforms.

By consolidating identity, authentication, and account management into a single platform, this module enables seamless access to all transport services through one trusted interface, reducing friction for travellers who currently rely on multiple systems such as Opal, toll accounts, and parking applications. Users access the platform through Service NSW, which acts as the trusted identity provider and primary authentication layer. Upon successful login, users are directed to their unified Mobility Account, where they can link and manage multiple transport services including MyE-Toll, Opal, Transurban (Linkt), etc. from a single interface. Service NSW handles credential verification, removing the need for separate logins across services. Once authenticated, the platform aggregates each linked service under one account, providing a seamless, consolidated view of the user's transport interactions and reducing fragmentation across the NSW mobility ecosystem.

Core Functions

1. **Create and manage a unique Mobility ID.** The platform creates and maintains a unique Mobility ID for each registered user, enabling a unified account across all mobility services. Users log in through Service NSW, which serves as a trusted identity provider, where their identity has already been verified via the Service NSW application. This ensures secure and reliable authentication before access to the platform is granted. The system also supports the management of linked identifiers for vehicles and authorised devices, ensuring that all relevant mobility assets are securely connected to the correct account.
2. **Maintain user profile and linked entities.** The platform stores and updates profile information, including personal details, linked vehicles, family groups, and authorised devices, using a relational database (RDB) structure.

The following entities form the core data model supporting Module 1. Each entity represents a distinct concept within the system, and their relationships are captured in the Entity-Relationship Diagram (ERD).

Table 5. ERD of Mobility Account Platform

Identity & Account

#	Entity	Description
1	MobilityAccount	Represents the primary digital identity of a registered resident. It serves as the central record linking mobility services, vehicles, devices, and entitlements.
2	AccountStatus	Reference entity defining the lifecycle state of a MobilityAccount (e.g., Active, Suspended, Deactivated).

Family Management

#	Entity	Description
1	FamilyGroup	Represents a collection of linked accounts for shared mobility usage (e.g., shared billing and access).
2	FamilyGroupMember	Records the association between a MobilityAccount and a FamilyGroup.
3	GroupStatus	Reference entity defining the current state of a FamilyGroup (e.g., Active, Suspended, Closed).
4	MemberRole	Defines the role of a member within a FamilyGroup (e.g., Owner, Member).
5	RelationshipType	Classifies the relationship between members (e.g., Parent, Child, Guardian).
6	MembershipStatus	Tracks participation status within a FamilyGroup (e.g., Invited, Active, Removed).

Business Management

#	Entity	Description
1	BusinessAccount	Represents a registered organisation managing fleets, billing, and mobility services.
2	FleetVehicle	Represents the association between a Vehicle and a BusinessAccount, enabling fleet management without duplicating vehicle records.
3	BusinessAccountMember	Records the association between a MobilityAccount and a BusinessAccount, defining authorised users.
4	BusinessRole	Defines the role within a BusinessAccount (e.g., Manager, Driver, Admin).
5	BusinessStatus	Reference entity defining the state of a BusinessAccount (e.g., Active, Suspended, Closed).

Vehicle Management

#	Entity	Description
1	Vehicle	Represents a registered vehicle linked to accounts for toll, parking, and mobility event attribution.
2	MobilityAccountVehicle	Represents vehicles that are owned or registered by a MobilityAccount
3	VehicleType	Reference entity classifying vehicle categories (e.g., Passenger Car, Motorcycle, Heavy Vehicle).
4	Make	Reference entity identifying the vehicle manufacturer or brand.
5	Colour	Reference entity capturing the registered vehicle colour.
6	OwnershipStatus	Defines the ownership or usage relationship (e.g., Registered Owner, Authorised User).

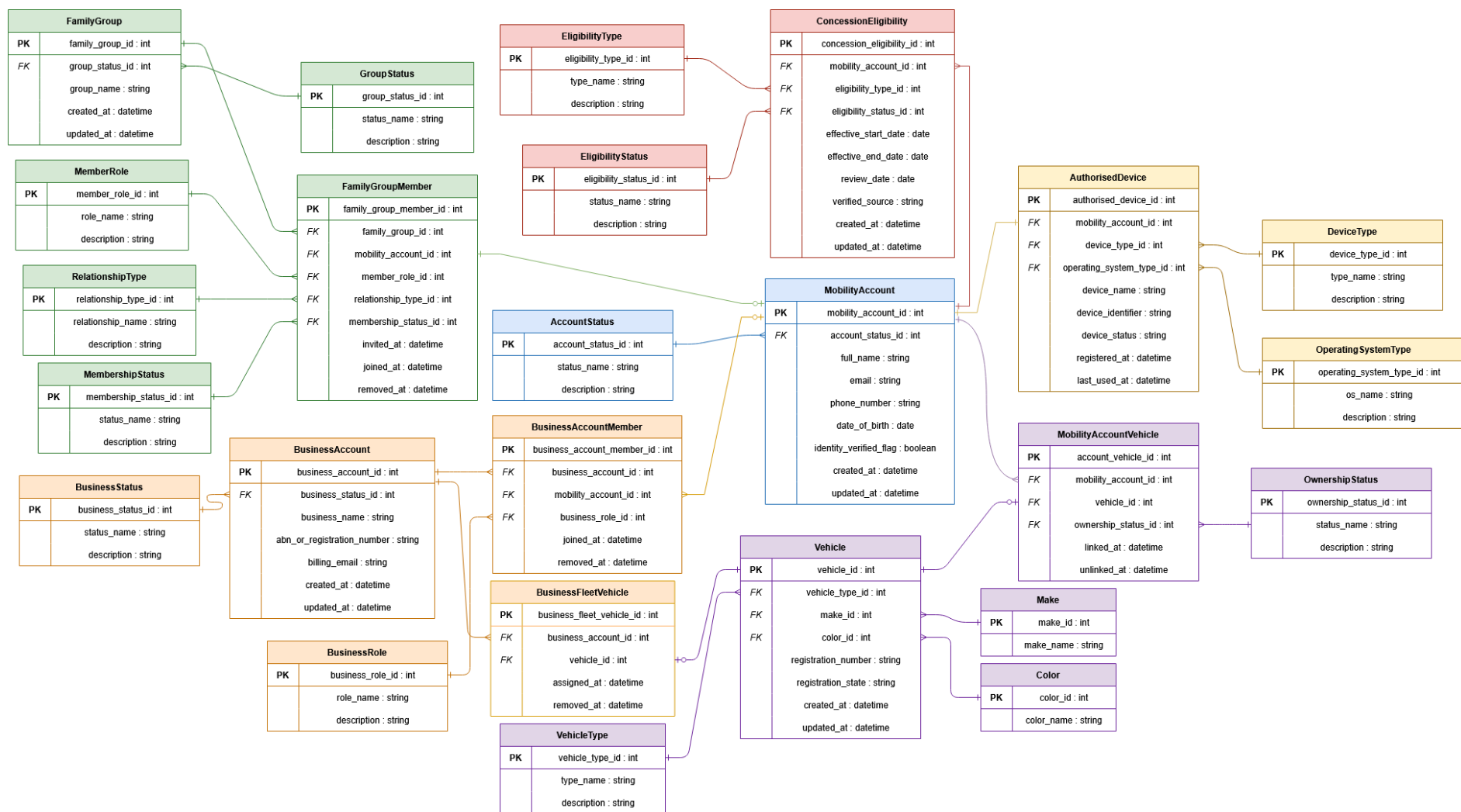
Device Management

#	Entity	Description
1	AuthorisedDevice	Represents a trusted device linked to a MobilityAccount for authentication and event capture.
2	DeviceType	Reference entity classifying device categories (e.g., Smartphone, Tablet).
3	OperatingSystemType	Reference entity identifying device operating systems (e.g., iOS, Android).

Concession System

#	Entity	Description
1	ConcessionEligibility	Records concession entitlements linked to a MobilityAccount for pricing adjustments.
2	EligibilityStatus	Reference entity reflecting concession status (e.g., Active, Expired, Under Review).
3	EligibilityType	Reference entity defining concession categories (e.g., Student, Senior, Pensioner, Disability).

Figure 2. Module 1 - ERD of SIMCP



- 3. Support role-based access control (RBAC).** The module defines what each user can see and do based on their role. Roles can be applied at the system level, family group level, and business level.

Table 6. RBAC and Description

System-Level Roles

#	Role	Description
1	System Administrator	Full technical control of the platform, including security settings, configuration, and system maintenance.
2	Administrator (Government / Platform Admin)	Manages user records, policies, concessions, and overall operational processes.

User Roles — Family Group Level

#	Role	Description
1	Family Owner	Creates and manages the family group, invites members, manages permissions, and links shared payment responsibility.
2	Family Member	Participates in the family group with permissions defined by the Family Owner.
3	Individual User	Uses the platform independently without belonging to a family group.

User Roles — Business Group Level

#	Role	Description
1	Business Owner	Creates and manages the business account, oversees billing and payments, manages members and roles, and maintains full control of fleet and account settings.
2	Business Manager	Manages fleet vehicles, including linking vehicles, assigning usage, and maintaining fleet records, with permissions defined by the Business Owner.
3	Admin Officer	Manages financial operations such as invoices, transactions, and payments, with restricted access to fleet operations.
4	Driver	Uses assigned vehicles with limited permissions, primarily for operational use and mobility tracking.
5	Individual User	Uses the platform independently without belonging to a business group.

- 4. Determine and maintain concession eligibility.** The platform stores, validates, and maintains concession eligibility for registered users in accordance with Transport for NSW policy. Eligibility is determined at account registration and updated when circumstances change, ensuring accurate fare discounts are applied across all linked services. The following concession types are supported:

Table 7. Concession Types and Description

#	Concession Type	Description
1	Child	Applies to passengers aged 4–15 years, eligible for discounted fares on Opal-integrated services.
2	Youth / Student (Under 16 on School Services)	Students travelling on approved school services may access free or reduced travel under eligible school transport programs.
3	Gold Opal (Senior / Pensioner)	Available to NSW Seniors Card holders and eligible Commonwealth Seniors Health Card holders, providing capped daily and weekly fares.

4	Concession (Commonwealth Concession Card Holders)	Available to holders of valid Commonwealth Concession Cards (e.g., JobSeeker, Disability Support Pension, Carer Payment), offering reduced fares across services.
5	Tertiary Student	Full-time students at approved tertiary institutions (e.g., university, TAFE) eligible for a Concession Opal Card, subject to enrolment verification.
6	Secondary Student	Full-time secondary students aged 16+ eligible for a School Opal Card, providing free or subsidised travel on eligible routes.
7	Vision Impairment / Disability	Individuals with qualifying vision impairment or disability receive concessional fares and may be exempt from certain charges.
8	Asylum Seekers	Eligible asylum seekers or bridging visa holders may receive concession fares under Transport for NSW eligibility rules.

Reference: <https://transportnsw.info/tickets-fares/eligibility-concessions>

- 5. Manage identity lifecycle.** The platform tracks account status throughout the identity lifecycle, including active, suspended, and deactivate.

Table 8. Managed Identity Lifecycle and Description

#	Status	Description
1	Pending	The account is not yet verified via Service NSW. Access to protected services is restricted until identity verification is completed.
2	Active	The account has been successfully verified and can access all authorised services.
3	Suspended	The account is temporarily restricted due to security concerns, policy breaches, or user request.
4	Deactivated	The account is permanently closed or disabled, and all access is removed.

- 6. Provide trusted identity data to other modules.** The module supplies secure identity tokens, account details, roles, and eligibility information to other modules so they can perform event capture, pricing, billing, and analytics using trusted data.

Inputs

Table 9. Module 1 - Solution Inputs

#	Input Name	Description
1	Authentication Token from Service NSW	OAuth 2.0/ OpenID Connect token returned after user login via Service NSW. Triggers account creation or session resumption and contains verified identity claims (e.g., name, DOB, customer number) without exposing credentials.
2	Verified Identity Profile	Verified identity data (e.g., full name, date of birth, identity level) provided by Service NSW, used to create or update the user's Mobility Account.
3	Vehicle Registration Data	Vehicle records retrieved via API from Service NSW and transport authorities, automatically linked to the user's account without manual input.
4	Concession Eligibility Data	Concession records (e.g., Seniors Card, Pension Card) sourced from Service NSW to determine eligibility type and status.

5	Service Linking Requests	User-initiated actions to connect external transport accounts (e.g., Opal, toll accounts) to the Mobility Account.
6	Device Registration Data	Device metadata (e.g., device type, OS) submitted by the mobile app upon first use to register authorised devices.

Outputs

Table 10. Module 1 - Solution Outputs

#	Output Name	Description
1	Mobility Account Record	Centralised account profile created or updated using verified identity data from Service NSW, serving as the foundation for all mobility interactions.
2	Platform Session Token	Secure, time-limited internal token issued after authentication, enabling authorised access across platform features and APIs.
3	Linked Service Registry	Record of all external transport services linked to the user's Mobility Account, enabling accurate charge attribution across modules.
4	User Profile Data	Structured dataset containing identity details, vehicles, devices, family groups, and roles, shared with downstream modules (Module 2–4).
5	Identity Lifecycle Status	Current account state (e.g., Active, Suspended, Deactivated), used across modules to control access and transaction eligibility.
6	Concession Status	Verified concession classification including eligibility type, effective dates, and review cycles, used by pricing module for discount application.
7	Authorised Device Registry	Verified list of registered user devices, supporting secure session management and accurate mobility event attribution.

Key Business Rules / Logic

Identity & Account

- Each MobilityAccount must have one current account status, stored using a status reference table such as AccountStatus.
- A MobilityAccount may be linked to multiple vehicles and multiple authorised devices.
- A MobilityAccount may belong to one family group as either a Family Owner or Family Member.
- A suspended or closed account must not be allowed to access protected services or generate new mobility transactions.
- Identity and access data shared with other modules must use secure tokens and approved APIs.
- All-important account events, such as login, verification, device registration, suspension, and role changes, will be recorded in an audit log.

Family Management

- A FamilyGroup must have exactly one Family Owner, but it can have many Family Members.
- Each FamilyGroupMember must have one MemberRole, such as Owner or Member, to define the user's permissions within the group.

Business Management

- A BusinessAccount may manage many fleet vehicles.
- A BusinessAccount may have multiple authorised members through BusinessAccountMember, allowing multiple individuals to access or manage the business account.
- Each BusinessAccountMember must link one MobilityAccount to one BusinessAccount and must be assigned one valid BusinessRole.

Vehicle Management

- Each Vehicle must have one valid vehicle type, make, and colour, stored using reference tables such as VehicleType, Make, and Color.

Device Management

- An authorised device must be linked to one valid account and may only be used if its status is active or trusted.
- Each AuthorisedDevice must have one valid device type and operating system type, stored using reference tables such as DeviceType and OperatingSystemType.
- A device linked to a suspended, deactivated, or closed MobilityAccount must not be used for authentication or mobility event capture.

Concession System

- Concession eligibility must be based on approved policy rules and trusted source data, not manual user claims alone.
- Concession status must include effective dates and review dates, because eligibility may expire or change over time.

Security, Audit & Governance

- Reference data such as AccountStatus, GroupStatus, BusinessStatus, MemberRole, BusinessRole, VehicleType, DeviceType, and EligibilityType will be centrally maintained to ensure consistency across the platform.

Interactions with Other Modules

Module 1 (Digital Identity & Mobility Account Platform) acts as the foundation layer of the SIMCP and interacts with all other modules by providing trusted identity, account, and access control information. It integrates with **Module 2** (Mobility Event Capture & Sensing Engine) by supplying verified user identities and authorised device information, allowing captured mobility events to be accurately linked to the correct user and account.

It also interacts with **Module 3** (Dynamic Pricing & Policy Engine) by providing user profile data, including concession status, account type (individual, family, or business), and relevant attributes required to apply pricing rules and policy adjustments. This ensures that pricing decisions are personalised and aligned with government policies.

For **Module 4** (Billing, Aggregation & Settlement System), Module 1 provides essential account identifiers, account ownership structures such as family groups and business accounts, and payment preferences, enabling accurate aggregation of charges and correct allocation of financial responsibility.

In addition, Module 1 supports **Module 5** (Analytics, Optimisation & Strategic Planning) by supplying structured and, where necessary, anonymised user and account data. This enables analysis of user behaviour, mobility patterns, and policy effectiveness while maintaining privacy and compliance requirements.

Overall, Module 1 functions as the central identity and access management layer, ensuring that all modules operate on consistent, secure, and trusted user information across the SIMCP ecosystem.

Implementation Technology (Indicative)

Cloud Platform: Amazon Web Services (AWS)

The SIMCP will be deployed on AWS as the primary cloud platform. AWS is selected because it supports the NSW Government's public cloud-first strategy and aligns with the existing architecture used by Service NSW, which acts as the trusted identity provider in this module. This ensures seamless integration with external identity services and APIs.

When adopting AWS as the cloud platform, it is recommended to primarily use AWS-native services to ensure seamless integration and efficient system design. Services such as Amazon Cognito for authentication, Amazon API Gateway for API management, and Amazon Aurora for data storage are designed to work together with minimal configuration, reducing development complexity.

In addition, AWS provides centralised security through AWS Identity and Access Management and encryption via AWS Key Management Service, ensuring consistent access control and data protection. AWS services also support automatic scalability and high availability, making them suitable for large-scale systems.

Reference: <https://versent.com.au/case-studies/versent-helps-service-nsw-digitise-and-secure-state-services-with-flexible-api-solution/>

Identity & Access Management (IAM): Amazon Cognito + AWS IAM

The platform will use Amazon Cognito together with AWS IAM to manage authentication and access control. Amazon Cognito is responsible for user identity management, including sign-up, login, multi-factor authentication, and integration with external identity providers such as Google or Service NSW via OAuth 2.0/OpenID Connect. IAM is used to control access between system components and enforce least-privilege permissions. This combination is chosen instead of building a custom authentication system because it significantly improves security, reduces development time, and ensures compliance with industry standards. Compared to Azure AD, Cognito integrates more seamlessly within AWS-native architectures and reduces cross-platform complexity. From a cost perspective, Cognito offers a free tier and scales based on active users, making it cost-efficient for a growing user base.

Reference: https://docs.aws.amazon.com/cognito/latest/developerguide/security_iam_service-with-iam.html

API Layer: Amazon API Gateway

The system will use as the main entry point for all client applications. API Gateway is responsible for handling incoming requests, routing them to backend services, enforcing security policies, and integrating with Cognito for authentication. It is chosen over self-managed API servers because it provides automatic scaling, built-in security features such as throttling and request validation, and seamless integration with other AWS services.

Reference: <https://aws.amazon.com/api-gateway/>

RDB: Amazon Aurora PostgreSQL

Structured and transactional data for SIMCP will be stored in Amazon Aurora PostgreSQL, a fully managed RDB service. Aurora PostgreSQL is selected because it provides high performance, strong consistency, and high availability with automatic failover, which are critical for managing sensitive data such as mobility accounts, vehicles, and user relationships. It is preferred over traditional self-managed PostgreSQL or MySQL due to its reduced maintenance, automatic backups, and better scalability. Compared to Azure SQL Database, Aurora offers better integration with other AWS services and improved performance for cloud-native workloads. While Aurora may have higher baseline costs than standard databases, its reliability and scalability justify the investment for a government-level platform.

Reference: <https://aws.amazon.com/rds/aurora/>

Encryption & Secrets Management: AWS Key Management Service (KMS) + AWS Secrets Manager

To secure sensitive data, SIMCP will utilise AWS KMS and AWS Secrets Manager. KMS is used to manage encryption keys for protecting data at rest and in transit, while Secrets Manager securely stores credentials such as API keys and database passwords. These services are selected over manual key management because they provide automated key rotation, fine-grained access control, and seamless integration with AWS services, significantly reducing security risks.

Reference: <https://docs.aws.amazon.com/secretsmanager/latest/userguide/security-encryption.html>

Authentication & Authorisation: OAuth 2.0 / OpenID Connect

Authentication and authorisation in SIMCP will follow industry standards using OAuth 2.0 and OpenID Connect, implemented through Amazon Cognito. These protocols enable secure, token-based authentication and allow users to log in using external identity providers such as Google or government services. This approach is preferred over traditional session-based authentication because it supports modern mobile and distributed architectures, improves scalability, and enhances security through token expiration and scope-based access control. Additionally, it allows seamless integration with third-party systems such as Service NSW, which is critical for a unified mobility platform. The cost impact is minimal as it is bundled within Cognito usage.

Governance & Compliance

Privacy Protection

Personal data collected by Module 1 is used only to set up and manage a user's Mobility Account. Only the minimum data needed is passed to other modules. Raw identity details are never shared with external operators. All data handling follows the NSW Privacy and Personal Information Protection Act and the Phase 0 Privacy Impact Assessment.

Security Controls

All users must log in with MFA. RBAC limits what each user can see and do. All data is encrypted in transit (TLS) and at rest (AWS KMS). Admin access to identity data is logged for review.

Audit and Traceability

Key account actions such as registration, device linking, concession changes, and account suspension are saved in an audit log with a timestamp and reason. This supports dispute handling and compliance checks without exposing personal data.

Regulatory Compliance

The module follows NSW privacy laws, government data breach rules, and Transport for NSW concession policy. Any changes to concession rules must go through the approval process set out in the Phase 0 governance framework before they take effect.

Summary

Module 1 is the identity layer of SIMCP. It gives each user one verified account that works across all transport services in NSW, removing the need to manage separate logins for Opal, tolls, parking, and other mobility services.

Users sign in through Service NSW, so their identity is already verified before they reach the platform. From there, the module links their vehicles, devices, concession status, and allows users to create group memberships and connect to external services into one profile that all other modules use.

By acting as the single source of truth for identity, Module 1 ensures that every trip captured, fare charged, and report produced across SIMCP is tied to the right person with the right entitlements. Security, privacy, and fair access are built into the module from the start.

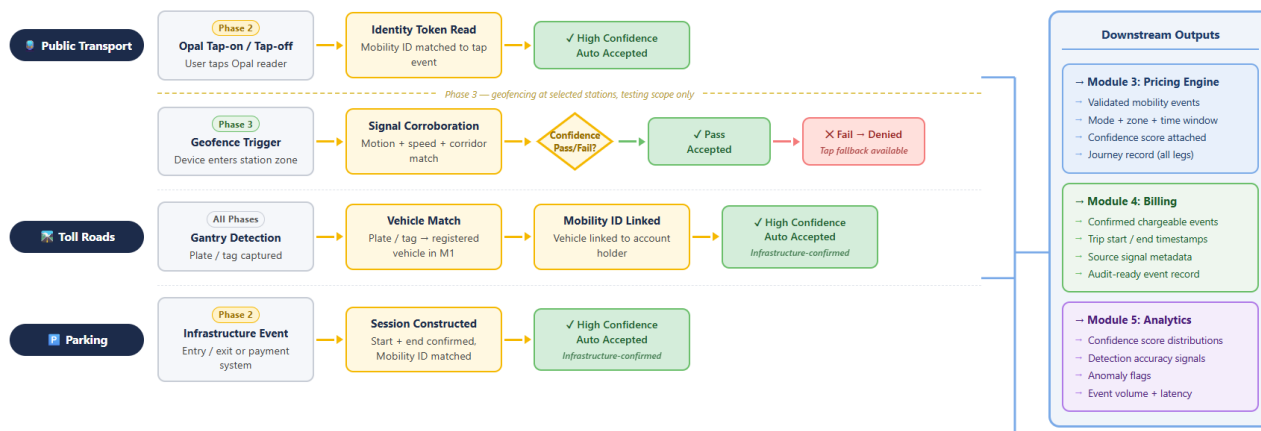
Module 2 — Mobility Event Capture & Sensing Engine

Module Overview

The Mobility Event Capture & Sensing is responsible for detecting and processing mobility events across three primary transport contexts: public transport, toll roads, and parking. Each context operates through a distinct detection mechanism with coverage and sophistication expanding progressively across delivery phases.

At a high level, the system operates in three stages: (1) interpreting raw signals into mobility events, (2) validating and ensuring their accuracy, and (3) enriching them into a consistent format suitable for pricing, billing, and analytics.

Figure 3. Module 3 - Solution Overview



Purpose

This Module serves as the core operational layer of the SIMCP, responsible for transforming raw mobility signals into structured, validated, and audit-ready events. While Module 1 (via Service NSW) establishes “who” the user is through a unified Mobility ID, this module determines “what” mobility activities are occurring across the transport ecosystem. It consolidates heterogeneous data sources into a unified, standardised event model.

Core Functions

End-to-end pipeline: detect → process → validate → enrich → publish

Table 11. Module 2 - Solution Core Functions

#	Function	Description	Output
1	Multi-Source Signal Ingestion (event-level)	Serves as the entry point by ingesting raw mobility signals from multiple sources, including transport infrastructure, authorised devices, and external providers. All inputs are validated, timestamp-aligned, and normalised into a unified schema, without interpreting mobility behaviour.	Standardised raw signals
2	Identity Binding and Attribution (event-level)	Resolves all incoming signals against a verified Mobility ID (via Module 1), ensuring each event is securely attributed to the correct user, vehicle, or authorised device. Events without successful identity resolution are not processed further (View Appendix 4A for Fall-back plan).	Identity-linked signals
3	Multi-Modal Event Detection (event-level)	Analyses identity-linked signals to detect discrete mobility activities across transport modes (e.g., trip start/end, mode detection, entry/exit points).	Atomic mobility events (time-stamped, mode-specific)

4	Signal Processing and Event Construction (journey-level)	Correlates and processes multi-source signals to construct structured mobility events, including single-leg and multi-leg journeys based on inferred or confirmed entry and exit points.	Structured mobility journeys (ordered, connected event sequences)
5	Event Validation	Ensures event accuracy by deduplicating signals, resolving conflicts (prioritising infrastructure data), validating logical sequences across all phases. In Phase 3, confidence scores to inferred events will be introduced for geofencing and route matching technology to support tap-free experience (View Appendix 4B).	Validated mobility events/journeys with confidence scoring
6	Event Enrichment and Standardisation	Enhances validated events with required attributes (e.g. mode, time classification, zone, duration) and transforms them into a standardised schema aligned with downstream pricing rules.	Enriched and standardised events
7	Real-Time Event Publishing	Publishes validated and enriched events in near real-time to downstream modules for pricing, billing, and analytics, while logging rejected events for audit purposes.	Published event streams + audit logs

Inputs

Table 12. Module 2 - Solution Inputs

Source	Input	Description
Module 1	Raw identity data (Mobility ID, authorised devices, registered vehicles)	Provides consented and verified identity references used to securely associate mobility signals with a user, device, or vehicle.
Public Transport System	Phase 2: Opal tap-on/tap-off transactions	Supplies confirmed, user-initiated transit events generated through fare interactions.
	Phase 3: Location-based transit events	Supports opt-in, location-assisted detection of boarding and alighting events without requiring physical taps.
	All-phases GPS location traces	Uses permission-based, context-limited location signals to support detection of mobility events and travel patterns.
External Mobility Systems	Toll data (gantry detection, plate/tag matching)	Provides verified vehicle movement events captured through existing tolling infrastructure.
	Parking events (entry/exit, session data)	Captures parking usage events, including duration and access points, with optional location-assisted detection in later phases.
Module 3 contextual data	Congestion, service disruptions, crowding indicators	Provides aggregated, non-personal contextual signals to improve event validation and detection accuracy.

Outputs

Outputs consist of validated, enriched, and standardised mobility events that are:

- Traceable (linked to Mobility ID, authorised devices, and source signals)
- Auditable (with full processing metadata and event lineage)
- Ready for downstream pricing, billing, and analytics systems

Each mobility event includes structured attributes such as:

- Event ID (unique identifier)
- Mobility ID / Account ID
- Trip / Journey ID
- Event type (e.g., trip start, in-progress, transfer, trip end)
- Transport mode
- Timestamp (start/end)
- Zone / corridor
- Detection method (e.g., tap, GPS inference, toll system)
- Source type (infrastructure, device, external system)
- Confidence score
- Processing timestamp

Events are produced in a standardised schema to ensure consistency across transport modes and interoperability with all downstream modules.

Table 13. Module 2 - Solution Outputs

Destination	Output	Description
Module 3 (Pricing Engine)	Validated mobility events	Enriched event records carrying transport mode, zone or corridor, time window classification, journey duration, and confidence score (Phase 3), ready for fare calculation and pricing rule application.
Module 3 (Pricing Engine)	Journey records	Consolidated multi-leg journey records with all legs linked under a single Journey ID, enabling accurate end-to-end fare calculation across transfer points.
Module 4 (Billing)	Confirmed chargeable events	Infrastructure-confirmed and high-confidence inferred events accepted by the pipeline, forming the factual basis for every charge generated in the billing cycle.
Module 4 (Billing)	Trip start/ end timestamps	Precise event timestamps used by Module 4 to allocate charges to the correct billing period and generate itemised statement records.
Module 5 (Analytics)	Confidence score distributions (Phase 3)	Aggregated confidence score data across modes, corridors, and detection methods, enabling Module 5 to monitor detection quality and identify drift or degradation.
Module 5 (Analytics)	Anomaly and fraud signals	Flagged events associated with impossible movement patterns, duplicate identity usage, or suspicious device behaviour, escalated for further investigation.
Module 5 (Analytics)	Event volume and latency metrics	Pipeline throughput and processing time data used to monitor performance against the ≤ 2 second SLA defined in Section 6.
Internal audit	Audit log records	Immutable records of all events processed by the pipeline, including accepted, held, and denied events, with reason codes and processing metadata for traceability and dispute resolution.
Account holder (via dispute path)	Denied event records	Records of low-confidence events excluded from billing, accessible to account holders through the dispute pathway for review and manual correction.

Key Business Rules / Logic

Table 14. Module 2 - Key Business Rules and Logics

Rules	Description
Identity and Attribution Rules	Every event must be linked to a valid Mobility ID from Module 1 Only authorised devices can generate valid mobility signals Vehicle-based events must match registered accounts
Event Detection Rules (Edge vs Cloud Split)	Edge devices may <i>only</i> trigger movement signals and geofence alerts Cloud system performs final mode classification and event confirmation No final event is accepted from edge alone
Event Construction Rules	A trip is initiated when sustained movement or transport entry is detected A trip ends after inactivity threshold or confirmed exit from transport network Transfers are defined by mode change within defined time window (e.g., 5–15 mins) Multi-leg journeys must be merged into a single journey record
Validation Consistency Rules	Duplicate signals within a defined time window must be removed Conflicting signals resolved using hierarchy: Infrastructure data (e.g., Opal / toll systems) > GPS inference Events must follow logical sequence: Start → In-progress → End
Confidence scoring rules (Phase 3 at testing scope)	Confidence is calculated using weighted signal reliability: • Tap / toll infrastructure → High • GPS route match, speed pattern, geofencing → Medium Events below confidence threshold are flagged or handled via fallback logic (View further details in Appendix 4B).
Event Enrichment Rules (Module 3 Alignment)	All events must include mode, time classification, zone/corridor, duration. Time must be mapped to peak / off-peak windows used in Module 3 pricing engine Zone definitions must align with pricing zones
Real-Time Processing Rules	Events must be processed in near real-time (stream-based) Only validated events are published downstream Each event must contain unique event ID, timestamp, confidence score (at Phase 3)
Fraud and anomaly rules	Detect impossible movement patterns (e.g., unrealistic speed in urban zones) Flag duplicate identity usage across locations Identify device mismatch or suspicious signal behaviour
Fallback and exception rules (View Appendix 4A)	Missing signals → infer using last known valid state Low confidence events → downgraded or excluded from pricing decisions System failure → revert to conservative rule-based classification

Interactions with Other Modules

Module 2 sits at the centre of the SIMCP data flow. It is the point where raw signals become trusted, structured events that every downstream module depends on.

Module 1 - Identity Supply. Before any event can be constructed, Module 2 draws on Module 1 to resolve the incoming signal against a verified Mobility ID. Every event that enters the pipeline is anchored to a confirmed identity, whether that is a person tapping an Opal card, a vehicle passing through a toll gantry, or a device triggering a geofence. Without this resolution, no event proceeds.

Module 3 - Bidirectional. The relationship with Module 3 runs in both directions. Module 2 supplies validated, enriched mobility events that form the factual input for fare calculation and pricing rule application. In return, Module 3 provides contextual signals including congestion levels, service disruptions, and crowding indicators that Module 2 uses to improve event validation accuracy and confidence scoring. Each module makes the other more intelligent.

Module 4 - Event Consumption. Every charge generated by Module 4 traces directly back to a validated event record produced by Module 2. Module 4 does not generate or modify events. It consumes them. This means the factual integrity of every billing statement depends entirely on the quality of what Module 2 publishes.

Module 5 - Aggregated Data Consumption. Module 5 subscribes to the event stream produced by Module 2 to monitor detection quality, track confidence score distributions, flag anomalies, and measure platform performance. The insights Module 5 produces and the governed recommendations it feeds back to pricing policy are only as reliable as the event data they are built on.

Implementation Technology (Indicative)

Coverage: event-driven + stream processing + modular integration.

Summary: Real-world mobility signals are first captured through edge and infrastructure systems, streamed via a real-time messaging layer, and processed using stream processing engines. Events then undergo validation, confidence scoring, and enrichment before being published as standardised mobility events to downstream systems including pricing, billing, and analytics platforms.

The following layers operate in a sequential event-driven pipeline.

Edge Sensing Layer

Mobility signals are captured at the edge through authorised mobile devices and transport infrastructure.

- Mobile devices use iOS Core Location / Core Motion and Android Location Services / Activity Recognition for GPS, motion detection, and geofencing triggers at defined station and stop boundaries.
- Infrastructure systems include Opal tap-on/tap-off, toll gantries, and parking systems (e.g., entry/exit detection and licence plate recognition).

Outcome: enables early detection of mobility activity close to the source.

Event Streaming Layer

Mobility signals are transmitted in real time to a central streaming platform.

Technologies: Apache Kafka / AWS Kinesis

Outcome: ensures scalable, fault-tolerant ingestion of high-volume mobility data.

Stream Processing Layer

Incoming events are processed in real time to transform raw signals into structured mobility events.

Technologies: AWS Lambda / Apache Flink / Kinesis Data Analytics

Functions include:

- signal correlation
- trip and transfer detection
- mode inference
- route matching against known transport corridors to infer the most likely service used.

Validation & Enrichment Layer

Events are validated and enriched for reliability and downstream usability.

Functions include:

- duplicate removal and consistency checks
- confidence scoring based on signal reliability
- enrichment with mode, time window, zone, and duration

Outcome: events log that is formatted to be suitable for pricing, billing and analytics.

Data Storage Layer

Validated events are stored for operational and analytical use.

- Operational storage: AWS DynamoDB (real-time access)
- Analytical storage: Data Lake (historical and analytics workloads)

API & Integration Layer

Secure APIs enable interoperability across modules and external systems.

Technology: AWS API Gateway / REST APIs

Supports integration with:

- Module 1 (Identity)
- Module 3 (Pricing Engine)
- Module 4 (Billing)
- external mobility providers

Monitoring Layer

System performance and data integrity are continuously monitored.

- AWS CloudWatch / logging systems

Covers:

- latency and throughput
- processing failures
- anomaly and fraud detection signals

Governance & Compliance

Data Minimisation and Purpose Limitation

The module collects only the minimum signal data necessary to detect, validate, and construct a mobility event. Device-based signals such as GPS traces and geofence triggers are used exclusively for trip detection and are not retained beyond the processing window unless required for dispute resolution or audit purposes. Raw signal data is never passed directly to downstream modules, ensuring only standardised, validated event records are published.

Privacy Protection

The collection of location-based signals from authorised mobile devices in Phase 3 requires explicit user consent, managed through the account registration process in Module 1. Users are informed of the nature, purpose, and scope of location data collection before any device-based sensing is activated. All location data is handled *in accordance with the Privacy Impact Assessment approved prior to Phase 3 deployment*, as defined in the Phase 0 governance requirements.

Audit and Traceability

Every event processed by the module, whether accepted, held, or denied, is recorded in an immutable audit log with a documented reason code, processing timestamp, detection method, and confidence score. This ensures full traceability from raw signal to published event and supports dispute resolution, regulatory review, and post-incident investigation without requiring access to raw personal data.

Anomaly signals escalated to Module 5 are subject to *human review* before any account-level action is taken to ensure that automated detection does not result in automated consequences that affect account holders.

Human Oversight

The module does not make autonomous charging decisions. All events published downstream carry a confidence score and detection method, ensuring that Module 3 and Module 4 have full visibility of the evidential basis for each event before pricing or billing is applied. Denied events are flagged for potential human review through the dispute pathway, and anomaly signals are escalated to Module 5 rather than acted upon automatically.

Security Controls

All data in transit between the edge sensing layer, streaming layer, and downstream modules is encrypted using TLS. Access to event data is governed by RBAC aligned to the platform-wide security baseline defined in Section 6. Device authentication is managed through the Authorised Device Registry in Module 1, ensuring that only registered and verified devices can generate valid mobility signals.

Regulatory Compliance

The module's event detection and data handling practices are aligned to the NSW Privacy and Personal Information Protection Act, the mandatory data breach notification requirements applicable to government platforms, and the platform-wide data governance standards defined in the SIMCP program governance framework.

Summary

Module 2 is the sensory foundation of SIMCP, transforming raw mobility signals from infrastructure systems, authorised devices, and external providers into validated, standardised event records that all downstream modules depend on.

The module operates through a sequential pipeline from edge signal capture through streaming, processing, validation, enrichment, and publication. This ensures only trusted, high-quality events reach Module 3, Module 4, and Module 5. Confidence scoring provides a transparent mechanism for managing detection uncertainty, while fallback rules and a dispute pathway protect account holders from erroneous charges.

Detection capability scales progressively across phases with Phase 2 establishes the core pipeline using existing tap and infrastructure signals, while Phase 3 introduces geofencing at selected stations on a controlled testing basis. Throughout all phases, Module 2 ensures that every charging decision made downstream is grounded in verified evidence, traceable to a confirmed identity and a documented confidence level.

Module 3 — Dynamic Pricing & Policy Engine

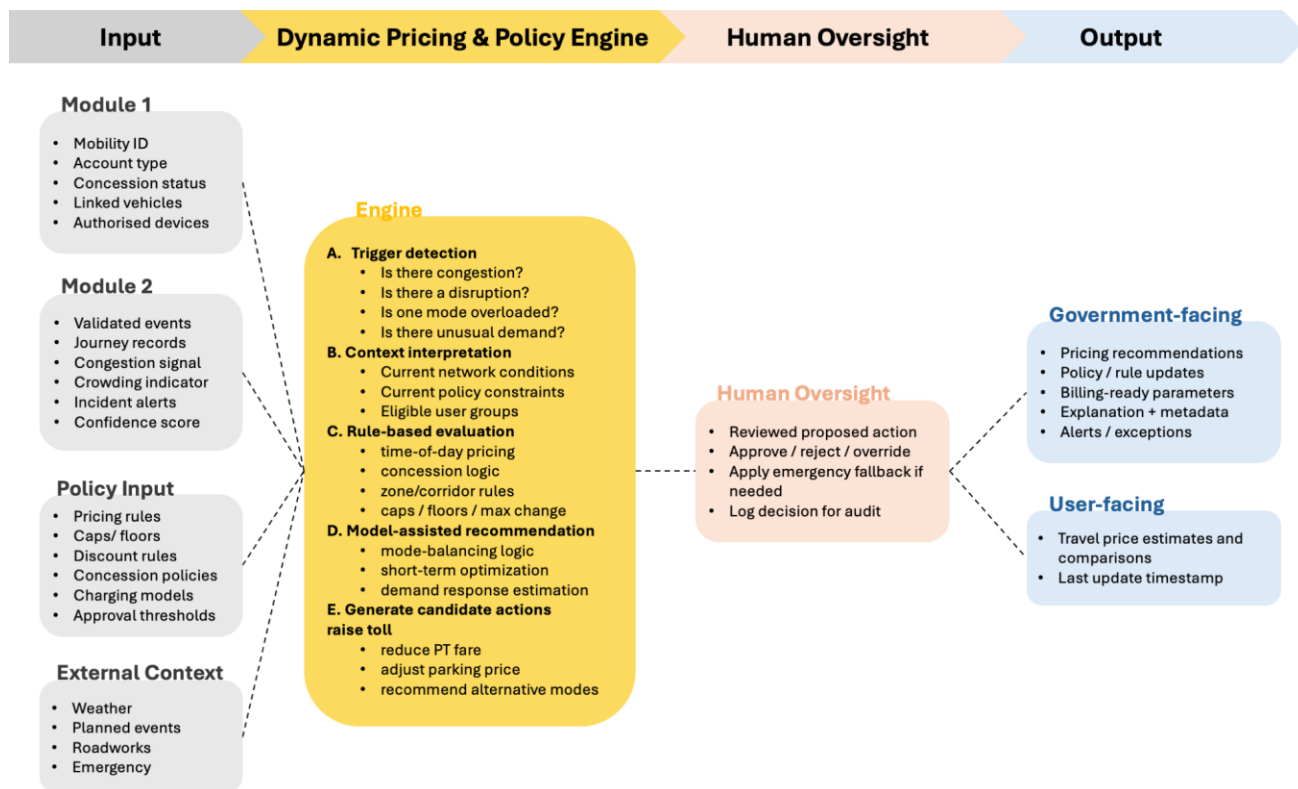
Purpose

The dynamic pricing & policy engine turns real-time mobility events into actionable pricing decisions. Its role is to apply pricing and policy logic to what's happening right now — for example sudden congestion, incidents, unexpected crowding on rail, or localised demand spikes. The goal is to recommend (and if authorised, apply) short-term pricing adjustments that help the government shift demand across modes, stabilise network performance, and balance revenue and cost impacts within agreed policy constraints.

This module focuses on real-time decisioning and operational control — turning live mobility events into actionable pricing recommendations within approved policy guardrails. Longer-term analytics, forecasting, and strategic policy refinement are covered in Module 5.

The following figure summarises the end-to-end decision flow for Module 3, highlighting how real-time events and policy inputs are translated into pricing recommendations under human oversight.

Figure 4. Module 3 - System Process of Dynamic Pricing



Note: Billing and settlement execution are handled in Module 4. Module 3 outputs billing-ready parameters and versioned policy updates only.

Core Functions

Core functions of the Dynamic Pricing & Policy Engine include:

Table 15. Module 3 - Solution Core Functions

#	Function	Description
1	Real-time congestion pricing	Adjust pricing in response to live congestion levels, incidents, and disruption signals.
2	Time-of-day adjustment	Recommend pricing signals that encourage travellers to shift modes or routes when one mode is overloaded (e.g., rail crowding → incentives for alternative modes/routes).
3	Environmental incentives	Apply policy-approved incentives or charges aligned to sustainability objectives.
4	Concession and eligibility adjustments	Apply concessions and eligibility-based discounts correctly and consistently, aligned to defined policy rules.

5	Real-time optimisation recommendations	Generate the “best available” pricing actions within guardrails (e.g., caps/floors, fairness constraints, operational limits).
6	Multi-modal price preview	Provide in-app price estimates across available modes (e.g., train, bus, driving, ride-share, bike-share where applicable) and highlight whether the current estimate is higher / lower / similar compared to the user’s usual baseline to support demand spreading during peaks or disruptions.

Inputs

Table 16. Module 3 – Solution Inputs

Source	Input/Description
Module 1	Mobility ID / Account ID and account ownership type (individual, family, organisation)
Module 1	Concession eligibility and user classification
Module 1	Linked vehicles, authorised devices, and relevant account relationships (where required for charging context)
Module 2	Standardised, validated mobility events across modes (public transport, tolls, parking; Phase 2+ shared mobility where applicable)
Module 2	Journey/event records with key attributes: timestamps (start/end), mode, zone/corridor, detection method, and confidence score
Module 2	Published operational signals used for real-time decisioning (e.g., congestion indices, crowding indicators, incident/disruption alerts)
Pricing and policy systems	Fare and pricing rules (by time window, zone, mode, and event type)
Pricing and policy systems	Discounts, caps/floors, concession policies, and eligibility conditions
Pricing and policy systems	Service-specific charging models and parameters (e.g., toll schedules, parking tariffs, incentive budgets)
Pricing and policy systems	Rule activation schedules, approval thresholds, and policy guardrails (e.g., max change frequency, escalation thresholds)
From external context sources	Weather conditions and severe weather alerts
From external context sources	Planned events and roadworks (demand drivers)
From external context sources	Emergency notifications that materially affect network operations

Outputs

Government-facing

Table 17. Module 3 – Government-Facing Outputs

Category	Destination	Output
Pricing Recommendations	STA + Councils + Operators (as advisory output)	Recommended toll changes (by zone/time)
		Fare incentives or targeted discounts for specific modes/zones
		Parking price adjustments (where policy allows)
Deployable Rules Updates	Policy Governance / Pricing Control (STA) + Module 4 + Operators	A versioned rule set with start/end times, target zones, and constraints
Decision explanation + audit metadata	Governance & Audit (STA) + Support/Disputes Team + Compliance	Trigger reason (e.g., incident type, congestion threshold), data inputs used, rule/model version, decision timestamp)

Alerts and Exceptions	Operations / Control Centre + Human Oversight Queue (Approvers) + Incident Response	Outlier pricing suggestions, policy boundary breaches, or situations requiring human override.
Billing-ready pricing parameters	Module 4	The correct fare/toll/discount parameters to apply per event/trip

User-facing

Table 18. Module 3 - User-Facing Outputs

Destination	Output
Account holder	A real-time estimated price range for each travel option, based on current policy rules and available data
Account holder	A comparison indicator showing whether the current estimate is higher / lower / similar relative to the user's baseline (e.g., typical time/day pattern)

Key Business Rules / Logic

Engine Development & Deployment

The pricing engine uses a rule-first, model-assisted approach. In early delivery, pre-approved policy rules provide safe operating boundaries (caps, floors, frequency limits, concession protections). Within those guardrails, model components are used to recommend the “best available” pricing actions for real-time conditions.

Engine build approach (offline)

Historical mobility events (tap-on/off volumes, toll transactions, parking sessions), network condition signals (congestion indices, incidents, service disruptions), and contextual data (weather/event flags) are used to train and calibrate models that estimate demand sensitivity and likely mode-shift responses. Outputs are evaluated using back-testing against past periods to confirm stability, fairness impacts, and budget/revenue constraints.

Engine serving approach (online)

In real time, the system ingests new events, derives features (e.g., crowding level, congestion severity, disruption duration), and runs a low-latency inference step to score candidate actions. The final decision remains constrained by policy rules and is subject to human oversight for high-impact changes.

Fallback / Safe mode (online)

If real-time signals are missing, delayed, or model health checks fail, the engine falls back to conservative, pre-approved rule sets (e.g., time-of-day pricing and fixed caps). For high-severity incidents, an emergency “safe mode” can freeze dynamic adjustments and require manual approval before any pricing changes are published. All fallback activations are logged for audit and post-incident review.

Continuous improvement

Model performance is monitored (accuracy, drift, false positives, equity metrics). Updates follow a controlled release process with versioning, approvals, and rollback to maintain auditability.

Interactions with Other Modules

Module 1 — Identity & Mobility Account

Module 3 retrieves Mobility ID and account context (e.g., account type and concession eligibility) to ensure pricing rules, discounts, and caps are applied consistently and fairly. Identity and entitlement data remains authoritative in Module 1 and is consumed by Module 3 only as needed for decisioning.

Module 2 — Mobility Event Capture & Sensing Engine

Module 2 is the primary event source for Module 3. It publishes validated, standardised mobility events and real-time operational signals (e.g., congestion, incidents, crowding, parking availability where available), which Module 3 uses to detect triggers and generate real-time pricing recommendations within policy guardrails.

Module 4 — Billing Aggregation & Settlement

Module 3 provides billing-ready pricing parameters (rule version, effective period, applied discounts/caps, and reason codes) to Module 4, enabling accurate charging, reconciliation, and dispute support. Module 4 is responsible for charge execution and statement generation, while Module 3 focuses on decisioning and governance.

Module 5 — Analytics, Optimisation & Strategic Planning

Module 5 provides longer-term insights (e.g., forecasting and policy simulation) that inform how pricing settings evolve over time. Module 3 remains focused on real-time operational control, while drawing on selected baselines or calibrated parameters from Module 5 to improve recommendation quality where appropriate.

Implementation Technology (Indicative)

The Dynamic Pricing & Policy Engine is implemented on AWS using an event-driven architecture. The design aligns directly to the end-to-end decision flow (Inputs → Engine → Human Oversight → Outputs), supporting low latency decisioning, strong auditability, and controlled policy governance.

Input (Ingest + Prepare)

Receive validated mobility events and context signals, then assemble the real-time decision context.

- Event intake (from Module 2):
 - Amazon EventBridge routes standardised mobility events (validated events, journey records, congestion/crowding/incident signals) into the pricing decision pipeline.
 - Amazon Kinesis Data Streams is used where sustained high-throughput streaming is required.
 - Amazon SQS buffers spikes and supports retry handling so downstream services remain stable during peak loads.
- API intake (from Module 1 and policy sources):
 - Amazon API Gateway + AWS Lambda provide secure APIs to retrieve identity/account context (Mobility ID, concession status, linked vehicles/devices) and policy inputs where needed.
 - AWS IAM + AWS Secrets Manager enforce least-privilege access and secure credential management.
- Real-time feature preparation:
 - AWS Lambda performs lightweight enrichment/feature building (e.g., zone mapping, time windows, crowding severity buckets, “usual baseline” reference signals).
 - Amazon DynamoDB stores operational state for quick lookup (active rule version, current thresholds, most recent context snapshot by zone/corridor).

Dynamic Pricing & Policy Engine (Decisioning)

Detect triggers, apply rules, generate candidate actions, and produce explainable outputs.

- Trigger detection & context interpretation:
 - Trigger logic is applied to incoming events (e.g., congestion spike, disruption severity, unusual demand, mode overload) using Step Functions as the workflow orchestrator, with Lambda as the execution units.
- Rule-based evaluation (policy guardrails):

- A configurable rules layer applies time-of-day pricing, concession logic, zone/corridor rules, caps/floors, and maximum change limits.
- Rule configuration is stored in Amazon DynamoDB to support low-latency rule retrieval, with explicit versioning and rollback support.
- **Model-assisted recommendation (bounded optimisation):**
 - For model-assisted scoring, Amazon SageMaker is used for training, model registry/versioning, and real-time inference endpoints.
 - Outputs remain bounded by policy guardrails to ensure decisions are explainable and defensible in a government setting.
- **Auditability (decision ledger):**
 - All decisions are written to an append-only audit log, capturing the inputs used, rule/model version, timestamps, and reason codes. Records are stored in Amazon DynamoDB for operational traceability and archived to Amazon S3 for long-term retention and compliance.
 - AWS CloudTrail captures administrative change actions for traceability.

Human Oversight (Approval + Safe Mode)

Ensure accountable decision-making for high-impact changes and provide emergency controls.

- **Approval workflow (human-in-the-loop):**
 - AWS Step Functions manages the approval path (review → approve/reject/override → publish), enforcing maker-checker controls where required.
 - Approval thresholds and escalation rules are configured as part of policy governance.
- **Safe mode / fallback:**
 - When data quality drops, model health checks fail, or severe incidents occur, the system can activate a safe mode that freezes dynamic adjustments and reverts to conservative rule-only pricing.
 - Fallback activation and manual overrides are logged to the audit ledger for post-incident review.
- **Monitoring and alerts:**
 - Amazon CloudWatch tracks pipeline latency, queue depth, error rates, and abnormal pricing outputs.
 - Amazon SNS is used to notify relevant operational/governance teams for urgent approvals or anomalies.

Output (Publish + Downstream Handoff)

Publish pricing outcomes and decision metadata for both government-facing and user-facing consumption and provide billing-ready parameters to Module 4.

- **Government-facing outputs:**
 - Event publication (push): Pricing recommendations, policy/rule updates, billing-ready parameters, explanation metadata, and alerts/exceptions are published as events via Amazon EventBridge to downstream modules and authorised government/operator consumers for near real-time processing.
 - Administrative retrieval (non-user-facing, pull): Restricted administrative APIs are provided via Amazon API Gateway + AWS Lambda to support on-demand lookup (e.g., active rule/version inspection, decision explanation, audit trace retrieval). And access is controlled via RBAC and least-privilege permissions.
 - Governance controls: Outputs are versioned and time-bound (effective start/end), including rule/model version and reason codes, to support reconciliation, auditability, and dispute handling.
 - User-facing outputs (journey price estimates):
 - Journey pricing API (user-facing, pull): The system provides Point A → Point B travel price estimates and multi-modal comparisons (higher/lower/similar than usual) via a secure API layer using Amazon API Gateway + AWS Lambda.

- Transparency measures: Responses include a “last updated” timestamp and a clear price disclaimer (prices are indicative and may change based on real-time conditions and policy rules).
- **Billing Interface (Module 4):** The dynamic pricing & policy engine sends billing-ready pricing parameters such as rule version, effective period, applied discounts/caps, reason codes to billing system for charge calculation and billing/statement generation.

Governance & Compliance

Financial Governance

- Full audit trail of all pricing decisions, billing-impacting adjustments, and settlement-related outputs
- Reconciliation between source usage events, applied pricing rules, billing statements, and settlement records

Privacy Protection

- Secure handling of personally identifiable information (PII), including minimisation and purpose-limited use
- Controlled data sharing based on role, necessity, and approved use cases (e.g., operator vs government access)

Security

- Encryption of data in transit and at rest
- RBAC, strong authentication, and secure API integration with operators and payment systems

Regulatory Compliance

- Alignment with public sector data governance and assurance expectations (e.g., auditability, transparency, and accountability)
- Transparent charging outcomes via itemised statements and clear “reason codes” for pricing changes
- Support for dispute handling, investigation workflows, and record retention requirements

Model & Decision Governance

- Model versioning and traceability, all model-assisted recommendations record model version, inputs used, and decision context
- Model performance monitoring, track accuracy/quality metrics and drift indicators; trigger review when thresholds are breached
- Approval and override monitoring, track approval rate, override rate, and “safe mode” activations to ensure ongoing human control and governance confidence
- Controlled rollout and rollback, support staged deployment (pilot zones/time windows) and rapid rollback to conservative rule-only mode if required

Summary

Module 3 is the real-time policy and pricing “decision layer” of SIMCP. It consumes validated mobility events and context signals from the SIMCP event capture pipeline, applies government-approved policy rules and guardrails, and produces versioned pricing outcomes (fares, toll adjustments, parking tariffs, incentives, and concessions) that are explainable and audit-ready. Where authorised, the module can publish approved rule updates; otherwise, it provides recommendations for human review and controlled release. It also supports user-facing price transparency through multi-modal price estimates and baseline comparisons (higher/lower/similar than usual), with clear timestamps and disclaimers to manage pricing volatility. The engine is designed to be rule-first and model-assisted, with strong governance controls including approval workflows, safe-mode fallbacks, and ongoing monitoring of model quality (accuracy/drift) and operational decision metrics (approval and override rates).

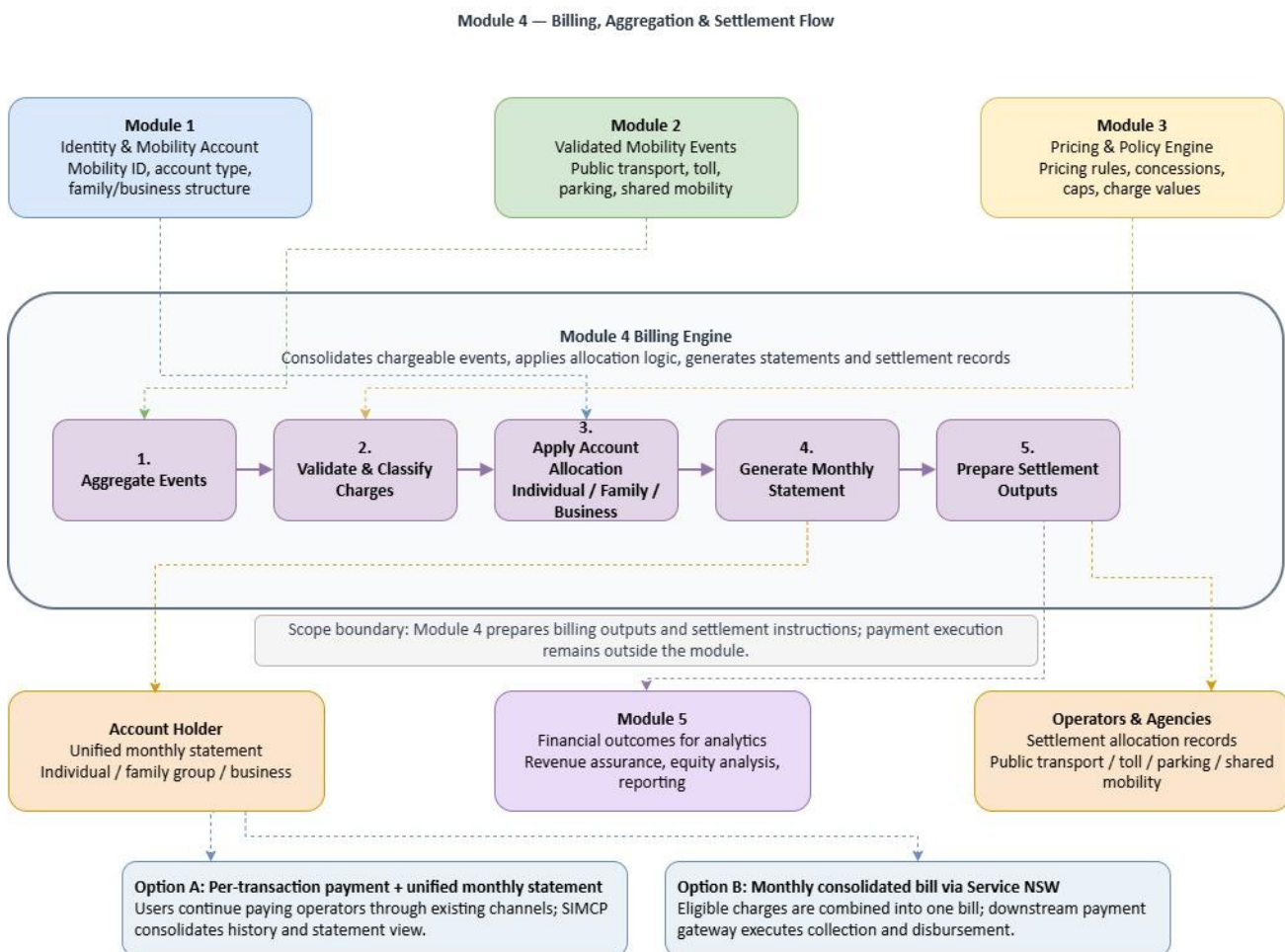
Module 4 — Billing, Aggregation & Settlement System

Purpose

The Billing, Aggregation & Settlement System serves as the financial orchestration layer of SIMCP, responsible for consolidating multi-modal mobility charges into a unified billing view and generating monthly statements for all account holders. It supports three account types – individual, family group, and business, enabling flexible charge attribution across a wide range of personal and organisational use cases.

This module **does not directly process payments or manage financial transactions**. Instead, it **prepares billing outputs and settlement instructions for downstream payment systems**, maintaining a clean separation between billing logic and payment execution. This separation improves scalability, security, and interoperability across the diverse range of operators and payment platforms participating in the SIMCP ecosystem.

Figure 5. Module 4 - Billing, Statement and Settlement Flow



Core Functions

Module 4 delivers seven core functions that together cover the full billing lifecycle — from event ingestion through to statement generation, operator settlement, and customer dispute resolution. The AI-powered billing assistant provides an additional customer-facing capability to support transparency and self-service.

Table 19. Module 4 - Core Billing Functions

#	Function	Description
1	Event aggregation & validation	All chargeable mobility events from Module 2 are ingested, deduplicated, and validated before billing is applied. Events are categorised by transport mode, service type, account ownership, and billing period.
2	Pricing application	Pricing rules, concession discounts, fare caps, and eligibility conditions from Module 3 are applied to each validated event, including time-of-day adjustments, zone-based pricing, and policy-driven incentives or surcharges.
3	Statement generation	All eligible charges within a billing cycle are consolidated into a single itemised monthly statement per account, with a full breakdown by date, time, transport mode, location, applied pricing rule, and any concessions applied.
4	Payment option support	Supports Option A (per-transaction payment with unified monthly statement) and Option B (monthly consolidated bill via ServiceNSW payment gateway). Both options consume the same billing output, ensuring consistent statement generation logic.
5	Settlement & revenue distribution	At close of each billing cycle, revenue allocation is calculated per operator and structured settlement outputs are generated for all participating operators. Payment execution remains outside module scope.
6	Dispute & adjustment handling	A structured four-step resolution process: Raise → Review → Adjust → Notify. All dispute records are retained for compliance and traceability.
7	AI-powered billing assistant	Embedded within the customer portal, enabling account holders to query charges, understand pricing decisions, and initiate disputes through natural language interaction.

Inputs

The module ingests eight input types:

Table 20. Module 4 - Billing Solution Inputs

Source	Input	Description
Module 1	User identity & account type	Account ID, ownership type (individual / family / organisation), concession status
Module 1	Linked vehicles & authorised devices	Required for attributing charges to correct account
Module 2	Validated mobility usage events	Public transport, toll, parking events with timestamps and service details
Module 3	Pricing rules & concession logic	Fare rules, caps, discounts applied per event/trip
Module 3	Billing-ready pricing parameters	Final charge values per event, ready for aggregation
External payment systems	Payment status updates	Success, failure, pending, chargeback notifications
External payment systems	Reconciliation / exception notifications	Mismatched totals, disputed transactions, settlement delays
Mobility operators (Phase 2)	Ride-share & bike-share usage events	Booking, trip start/end, service zone data

Outputs

The module produces seven structured outputs:

Table 21. Module 4 - Billing Solution Output

Destination	Output	Description
Account holder	Consolidated monthly billing statement	Single itemised statement covering all mobility charges for the billing cycle
Account holder	Itemised charge records	Per-event breakdown with timestamp, service category, and amount
External payment systems	Billing output / payment instructions	Structured billing data for payment execution
Participating operators	Settlement allocation records	Revenue distribution instructions per operator (transport, toll, parking, etc.)
Government agencies	Reporting outputs	Financial summaries for oversight and compliance
Module 5	Financial outcomes	Aggregated billing and settlement data for analytics and revenue assurance
Internal audit	Audit logs & dispute records	Full traceability of all billing events, adjustments, and resolutions

Key Business Rules / Logic

Billing Account Structure. SIMCP supports three billing account types, each with distinct attribution, statement, and payment responsibility rules:

Table 22. Module 4 - SIMCP Billing Account Structure

Billing Type	Event Attribution	Statement Recipient	Payment Responsibility
Individual	Linked Mobility ID	Account holder	Account holder
Family Group	Per member, individually tracked	Family Owner	Family Owner
Business / Fleet	Per employee or vehicle	Organisation account	Fleet Manager / Business Owner

Individual Billing. All chargeable events linked to an individual Mobility ID within a billing cycle are consolidated into a single monthly statement addressed to that account holder. Concession discounts, fare caps, and policy-based incentives are applied as received from Module 3. The account holder is responsible for payment via their registered payment method.

Family Group Billing. Within a family group, each member's mobility events are individually tracked and attributed. The Family Owner is the nominated billing responsible party and receives a consolidated statement covering all family members' charges, with an itemised breakdown by member. Individual members may optionally receive a personal summary view. Payment responsibility rests with the Family Owner. Family group billing is limited to a maximum number of members per group, as defined by platform policy, and group configuration is managed via the Family Owner role defined in Module 1.

Business and Fleet Billing. Business accounts support charge attribution for employees and linked fleet vehicles. Charges generated by an employee using a linked vehicle or an authorised device are attributed to the business account and appear on the organisation's monthly statement, with a breakdown by employee or vehicle as configured by the Fleet Manager

role defined in Module 1. Mixed-use scenarios, where an employee uses a fleet vehicle for personal travel, are handled through rule-based allocation configured at the account level.

Settlement Allocation Logic. Revenue collected through the unified billing statement is distributed to participating operators based on each operator's attributed usage events within the billing cycle, using the service provider identifier recorded at event ingestion (Module 2). The billing engine calculates each operator's share of total revenue and generates structured settlement allocation records specifying the payable amount per operator, the applicable billing period, and any adjustments. Payment execution is handled by external financial systems — Module 4 produces the settlement instructions only.

Scope Boundary. It is important to note the scope boundary for business billing: SIMCP supports individual employee mobility billing attributed to a linked organisational account. Full corporate fleet billing integration, including replacement of existing operator-direct arrangements for fleet tolling and heavy vehicle schemes, is outside the scope of Phase 1 and Phase 2 of the current program. Businesses with existing Transurban or Linkt fleet accounts may continue those arrangements independently. SIMCP business billing is designed to complement, not replace, existing corporate mobility arrangements in the current program phases.

Phase-Based Service Expansion. In alignment with the Section 9 delivery roadmap, billing coverage expands progressively across phases:

- **Phase 1 (Foundation Build)** establishes the billing baseline covering public transport, tolling, and parking, with statement generation and reconciliation capabilities in place.
- **Phase 2 (Greater Sydney MVP)** implements the agreed payment option end-to-end, including dispute handling and operator settlement for all in-scope Greater Sydney services.
- **Phase 3 (Statewide Expansion)** onboards shared mobility providers such as ride-share and bike-share, beginning with booking and receipt integration before progressing to full billing integration where commercial partnerships allow, and expands coverage to all NSW regions.

Interactions with Other Modules

Module 4 sits at the financial convergence point of the SIMCP data flow, drawing on outputs from three upstream modules and feeding into the analytics layer downstream.

Module 1 provides the account structure, ownership hierarchy, and payment preferences that govern how charges are attributed and statements are addressed across all account types.

Module 2 supplies the validated usage events that form the factual basis for every charge in the system.

Module 3 provides the pricing parameters — fares, tolls, discounts, and caps — applied to each event before billing is calculated. External payment systems receive billing outputs and execute payment collection independently of this module, while participating operators receive settlement and reconciliation outputs for revenue reporting.

Module 5 consumes billing outcomes as a key input for revenue assurance monitoring, equity analysis, and financial reporting, and may raise anomaly flags back to Module 4 where billing patterns warrant investigation.

Implementation Technology (Indicative)

The module adopts a **scalable, event-driven architecture** built on **cloud-native components**:

- API-based integration connects the module to transport operators and external payment systems.
- Event-driven pipelines ingest and process validated mobility usage data from Module 2
- A rules-based billing engine handles charge aggregation, pricing application, and account allocation logic.

The module's technology stack is organised across four layers.

- For data storage, an immutable transaction ledger (implemented via DynamoDB) retains all charge records to support auditability and event replay, while an RDB stores billing statements, settlement records, and reconciliation outputs.
- For compute, serverless functions (Lambda) handle billing engine logic, pricing application, and statement generation.
- For messaging and orchestration, an event bus (EventBridge / SQS) decouples services and manages billing workflows, ensuring reliable processing under high transaction volumes.
- For reporting, dashboards and interfaces provide operators and government agencies with visibility into billing outcomes, settlement status, and exception queues.

Governance & Compliance

Financial governance is maintained through a full audit trail of all billing events, adjustments, and settlement outputs. Reconciliation is performed between source events from Module 2, applied pricing rules from Module 3, generated billing statements, and settlement records before each cycle is finalised. Any discrepancy beyond the defined tolerance threshold triggers an exception workflow with documented resolution prior to payment release.

Privacy protection is enforced through secure handling of PII, with data sharing strictly controlled by role and purpose. Only the minimum data necessary for billing and settlement is passed to downstream systems and external operators, in alignment with the privacy principles established in the platform-wide PIA.

Security controls include encryption of data in transit and at rest, RBAC across all billing functions, and secure API integration with operators and payment systems. These controls align with the platform-wide security baseline defined in Section 6.

Regulatory compliance is maintained through transparent, itemised billing statements with clear reason codes for any pricing adjustments, structured support for dispute handling and record retention requirements, and alignment with NSW public sector data governance standards applicable to financial systems. All billing outputs are designed to support audit and regulatory review without requiring access to raw personal data.

Summary

Module 4 provides the financial backbone of SIMCP, transforming validated mobility events and pricing parameters into consolidated monthly statements, operator settlements, and auditable financial records. It supports three account types: individual, family group, and business, with clearly defined billing logic for each, and is designed to accommodate two payment models that reflect the different ownership arrangements across government and private operators.

By orchestrating billing without directly executing payments, the module maintains a clean separation of concerns that supports scalability, interoperability, and regulatory compliance. Business and fleet billing is supported within defined scope boundaries, with full corporate fleet integration reserved for future phases. The phased expansion model ensures that new mobility services and operators can be onboarded incrementally – from the Greater Sydney MVP through to statewide coverage – without redesigning the core billing architecture.

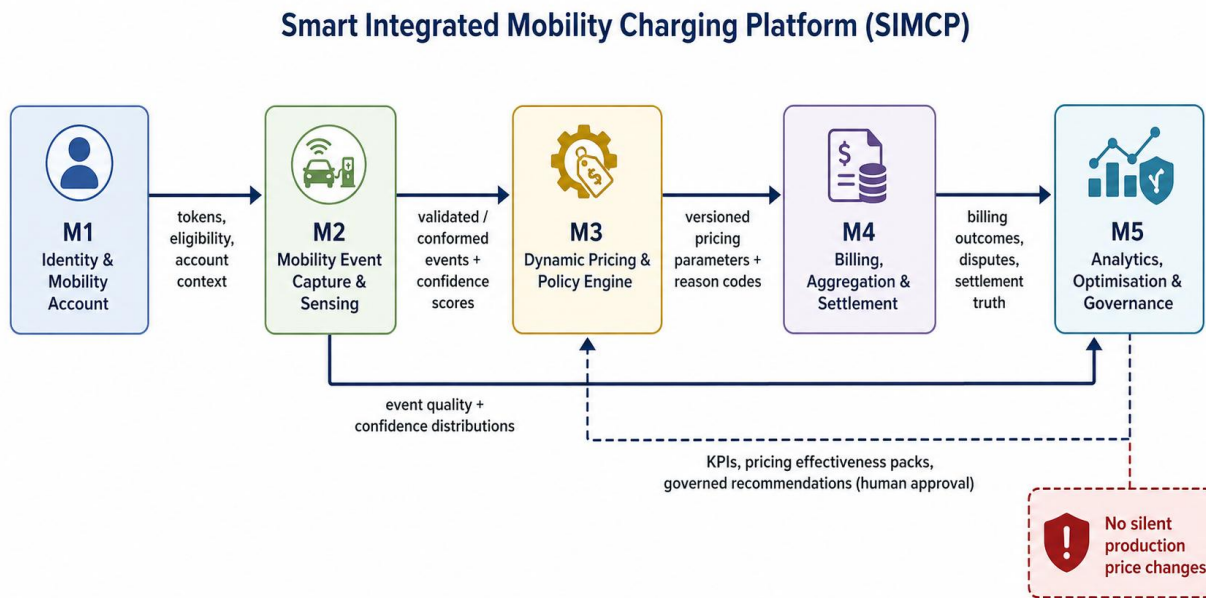
Module 5 — Analytics, Optimisation & Governance Platform

Purpose

The Analytics, Optimisation & Governance Platform is STA's authoritative environment for turning mobility transactions into evidence-based decisions. It consumes standardised usage and financial outcomes from across SIMCP, monitors policy effectiveness (congestion, mode shift, equity, revenue integrity), and supplies curated insight back to pricing and service design.

The platform is intentionally separated from operational charging paths so that analytics scale, experimentation, and reporting do not compromise real-time authorisation and billing performance.

Figure 6. Module 5 - High level enterprise architecture diagram for SIMCP



Core Functions

Table 23. Module 5 - Solution Core Functions

#	Function	Description
1	Medallion-style analytics foundation	Maintain an immutable raw landing zone for audit and replay; curated (cleansed, conformed) datasets; and consumption structures including an operational data store (ODS) for near-real-time monitoring, a data warehouse and subject-specific marts for official reporting, and optional governed data virtualisation across those stores
2	Confidence-score intelligence (Module 2 aligned)	Ingest and analyse confidence scores attached to conformed events (especially inferred / tap-free and GPS-route-matched trips). Monitor distributions, drift, and corridor- or mode-level patterns of low-confidence events; correlate with dispute rates, exclusions from rating, and manual review volumes; and produce actionable recommendations (for example tuning path-matching thresholds, geofence boundaries, or fallback rules) so capture quality improves without weakening privacy posture.
3	Reporting, dashboards, and transparency packs	Deliver executive and operational dashboards, self-service analytics for authorised agency and operator roles, KPI catalogues (usage, reliability, capture latency, settlement outcomes, mode share, geographic equity indicators), and structured reporting packs for transparency obligations.
4	Policy and network "what-if" analysis	Scenario modelling on aggregated and pseudonymised datasets, feeding recommendations only to Module 3 rather than embedding opaque autonomous optimisation inside production charging.
5	Pricing effectiveness & mode-shift feedback (feeds Module 3)	Produce versioned pricing effectiveness packs that compare outcomes before and after material policy changes, including mode-shift effects (modal share, peak spreading, equity cohorts). Where Module 3 adjusts suggested prices using Module 5 insight, Module 5 defines the metrics (for example incremental revenue, demand moved off overloaded corridors, unintended distributional shifts), cohort definitions, and reporting cadence so Module 3 consumes a stable, auditable signal.

6	Data quality, anomaly, and revenue assurance	Anomaly and drift alerts on ingestion, schema conformance, duplicate/late events, and financial reconciliation gaps; documented KPI definitions and lineage.
7	Curated outputs for government planning	Data extracts and APIs for accredited planning tools; archival datasets for longitudinal evaluation. Where public transparency is required, outputs are aggregated to minimise re-identification risk and published through STA-approved channels.

Inputs

Table 24. Module 5 - Solution Inputs

Source	Input	Description
Module 2	Conformed mobility events; trip/usage segments (timestamps, mode, zone/corridor, confidence scores, validation flags, linkage handles)	Provides validated and structured mobility data used as the primary input for downstream processing, with privacy-preserving linkage.
Module 3	Rated charge components; policy version identifiers	Supplies pricing outputs and applied rule versions to support billing consistency and traceability.
Module 4	Invoices, adjustments, settlement batches, exception flags	Provides billing outcomes and financial records for reconciliation, auditing, and exception handling.
STA Master Data	Network topology, zones, concession categories, calendar effects	Provides reference data required for consistent interpretation of mobility events and pricing logic.
Module 1	Entitlement and consent posture summaries	Supplies user permissions and eligibility data to ensure lawful reporting and data usage.
External / Integration Sources	Batch extracts and streaming feeds (e.g., toll systems, micromobility providers)	Supports both batch and real-time data ingestion from legacy and external systems via defined integration patterns.

Outputs

Outputs include published dashboards and reports; data extracts and APIs for accredited government planning tools; anomaly and drift alerts for data quality and revenue assurance; documented KPI definitions and lineage; pricing effectiveness and mode-shift briefings for Module 3 policy owners (supporting governed parameter updates rather than silent production changes); and archival datasets for longitudinal evaluation.

Key Business Rules / Logic

De-identification and re-identification risk controls

Datasets used for cross-agency policy simulation, external sharing, or published statistics must follow approved pseudonymisation and aggregation rules. Public-facing transparency packs must apply small-cell suppression, minimum cohort thresholds, and geographic/temporal generalisation (e.g., corridor/zone rollups and time-bucketing) so re-identification risk is reduced by design.

Role-based analytics access

Access is purpose-limited; raw PII paths are restricted to authorised roles and use cases.

Official KPIs from approved marts only

Published KPIs are calculated from versioned “gold” logic, so comparisons across policy periods remain reproducible.

Records and retention

Raw data retention, aggregation windows, and pseudonymisation follow STA records management and privacy schedules.

Human-in-the-loop for production impact

Any parameter or rule bundle recommended for Module 3 production deployment must pass human approval and change control; Module 5 does not silently alter charges.

Batch vs stream reconciliation

Conflicts between batch- and stream-derived metrics are resolved through defined reconciliation jobs and exception workflows.

Phased capability

Phase 0 Program Foundation (Set-up & Alignment)

Confirm scope, governance, target operating model, payment-model boundaries, and data-sharing principles before technical build; analytics activity is planning-only (KPI ownership, retention/pseudonymisation expectations, assurance hooks)

Phase 1 Foundation Build (Platform + Integration Baseline)

Baseline telemetry, data landing and conformance, core financial and operational dashboards, and foundational pricing-effectiveness views (post-hoc comparisons tied to policy versions).

Phase 2 Core Modes & Billing + Future Capability Build (MVP, Greater Sydney)

Near-real-time operational dashboards expanded KPI catalogue aligned to MVP modes, and structured feedback into Module 3 tuning cycles within governance guardrails.

Phase 3 Expansion & Optimisation + Frictionless Travel Rollout (NSW-wide)

Mature predictive and dynamic demand views where appropriate (traffic and mode-load forecasting, scenario libraries), deeper confidence-score analytics supporting tap-free scale-up, and continuous pricing-effectiveness reporting as new partners and modes onboard.

Interactions with Other Modules

Module 2 supplies the primary factual stream of “what happened,” including confidence metadata for inferred events.

Module 3 consumes analytics on congestion, mode share, equity impacts, and pricing effectiveness signals to tune rules within statutory bounds and publishes which policy version applied to each chargeable outcome.

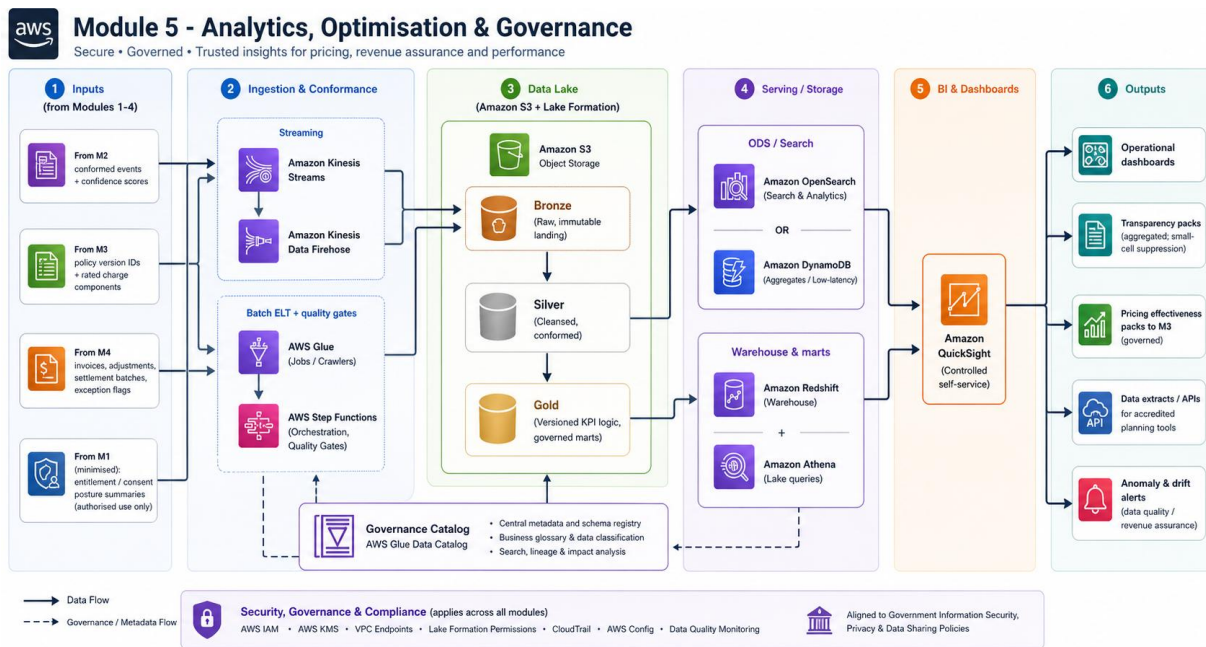
Module 4 provides financial truth for cash collection, allocation across operators, and refunds—analytics depends on these outcomes for revenue and fairness monitoring.

Module 1 provides the minimum necessary identity and eligibility context; analytics does not replace Module 1’s identity decisions but joins on controlled keys for authorised internal analysis only.

Implementation Technology (Indicative)

Indicative components include a cloud data lake for raw/curated zones, batch/stream conformance pipelines, an ODS for low-latency operational monitoring, a governed warehouse for official marts, and a Business Intelligence layer for dashboards and transparency packs.

Figure 7. Module 5 – AWS Cloud Solution for SIMCP



To remain consistent with the AWS-specific implementation style used elsewhere in this draft, an indicative AWS reference stack for Module 5 is:

Data lake governance and zone

Amazon S3 for raw/bronze-silver-gold storage, with AWS Lake Formation for permissions, cataloguing, and cross-domain governance (backed by AWS Glue Data Catalog).

Batch and streaming ingestion

Amazon Kinesis (Data Streams / Firehose) for streaming mobility events and outcomes; AWS Glue (and where needed, AWS Step Functions) for batch ELT orchestration and data quality gates.

Near-real-time ODS and search

Amazon OpenSearch Service (or DynamoDB for key operational aggregates) to power low-latency operational views and anomaly triage.

Warehouse and marts

Amazon Redshift for governed reporting marts, with Amazon Athena for lake-native ad hoc query where appropriate.

BI / dashboards

Amazon QuickSight for executive/operational dashboards and controlled self-service analytics.

Security and audit baseline

AWS IAM (role-based access), AWS KMS (encryption), AWS CloudTrail + CloudWatch for audit and observability; AWS Organizations guardrails aligned to STA governance.

This mapping is indicative and keeps procurement/vendor selection outside scope, while still making service choices concrete for implementation planning.

Tiered and on-demand service selection

To control cost during early phases, Module 5 is designed to be deployed in tiers. For Phase 1 and early Phase 2, STA can prioritise lake-native and pay-per-query patterns (e.g., Amazon S3 + AWS Glue + Amazon Athena) and stand up only the minimum ODS and dashboard footprint required for operational visibility. As data volume, concurrency, and reporting obligations grow, STA can progressively introduce a governed warehouse layer (e.g., Amazon Redshift) and additional performance-oriented services. Data retention and storage cost are managed through lifecycle policies on the data lake (e.g., automated tiering and archival for older raw/bronze data), aligned to STA records schedules.

Governance & Compliance

The platform operates under STA data governance: data owners for each domain, stewardship for quality rules, documented lineage from source events to published KPIs, and periodic independent review of sensitive use cases.

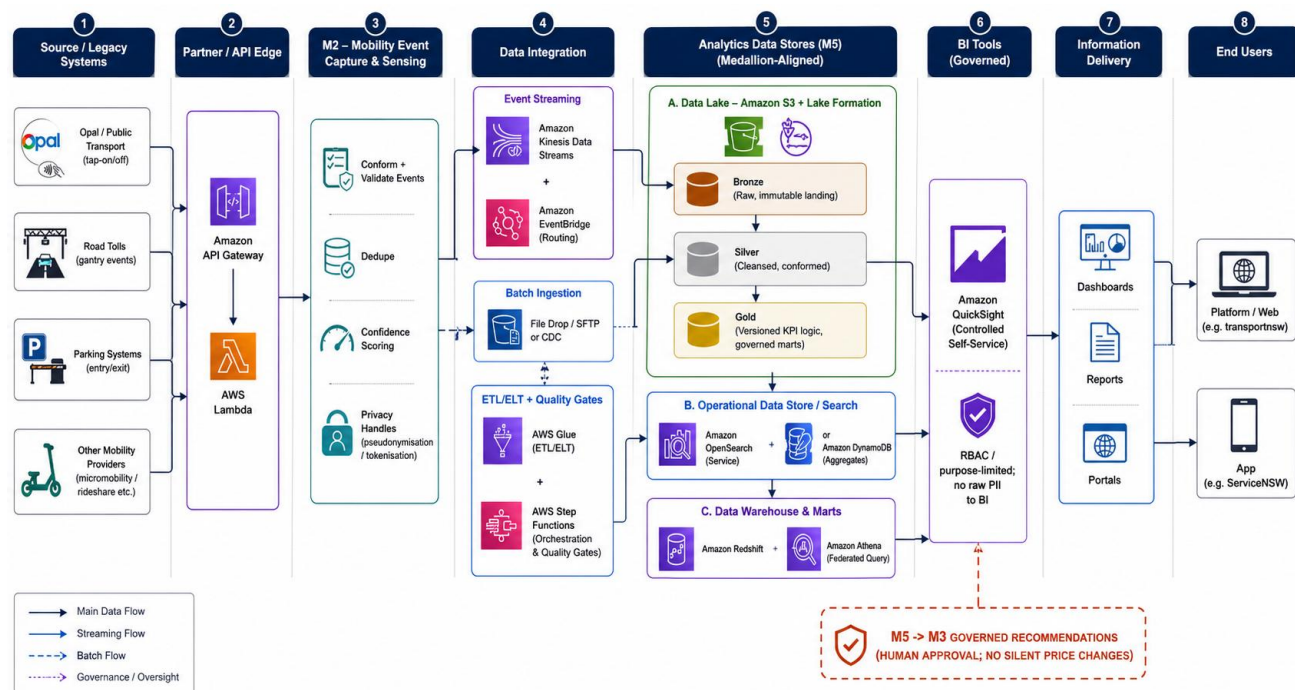
Privacy impact assessments cover new dashboards and external data sharing. Model risk management applies where predictive or optimisation aids are used: documented assumptions, monitoring bias against protected cohorts where legally relevant, and human accountability for decisions that affect charges or service levels.

Summary

Module 5 closes the loop between operation and policy by making SIMCP measurable, explainable, and improvable. It preserves separation of duties between charging operations and insight, while giving STA a durable analytics estate that can absorb new modes and partners — including tap-free capture at scale — without redesigning the core transaction path.

5. System Integration & Interoperability

Figure 8. System Integration and Interoperability Architecture Diagram for SIMCP



SIMCP integration is organised around a single logical journey — identity and eligibility, event capture, rating, financial aggregation, and insight, while accepting that legacy mobility systems were never built as one product.

The integration posture therefore combines an API-first edge for partner and real-time channels, an event-streaming backbone for high-volume, time-sensitive signals, and batch ingestion for systems that remain file- or extract-based. This hybrid pattern reduces forced “big bang” replacement of stable back offices while still allowing STA to converge on one mobility account narrative over time.

End-to-end flow begins in Module 1, where the resident or fleet obtains a mobility account, credentials, and entitlements. When travel or usage occurs, Module 2 ingests evidence from operator and infrastructure systems—Opal-class public transport, tolling, parking, registration-related charges where in scope, shared micromobility, and future modes, normalising them toward a common mobility event envelope. Module 3 rates each conformed event (or journey aggregate) using policy rules, time windows, and network state, producing a priced outcome bound to a policy version. Module 4 performs billing orchestration, presentment, payment capture, allocation, and settlement, including exception handling for disputes and partial failures. Module 5 subscribes to conformed events and financial outcomes to monitor performance, fairness, and reliability, and feeds approved recommendations back to policy owners rather than silently altering charges. The API gateway provides external-facing REST (and optionally GraphQL) interfaces for onboarding, configuration, and query patterns that suit partner IT estates, backed by OAuth2/OIDC-style security aligned to Module 1.

Where concrete AWS services are used for this integration spine (consistent with other modules), a typical mapping is: Amazon API Gateway + AWS Lambda for partner-facing APIs; Amazon EventBridge for domain event routing; Amazon Kinesis for high-volume streaming ingestion; and Amazon S3 for batch landing zones and replay-able raw evidence archives, with schema enforcement and contract tests embedded in the continuous integration and deployment (CI/CD) pipeline.

High-volume ingestion for toll-like and micromobility-like streams is directed to the event streaming layer, where topics are partitioned by operator or geography to contain blast radius and aid scaling. Batch interfaces—secure managed file exchange, database CDC where agreed, or scheduled extracts—land in a controlled ingestion zone with checksum, virus scanning, and schema validation before promotion to analytics and operational pipelines.

A common mobility event schema is the contract between producers and SIMCP. At minimum it carries a stable event identifier, event type, timestamp(s), location semantics agreed per mode (zone, corridor, stop, or geohash precision as policy allows), mode and service identifiers, a pseudonymous traveller or vehicle handle linked only inside STA’s trust domain, fare or toll product hints where supplied, and data-quality flags.

Producers may send source-specific extensions in namespace fields, but the canonical core is validated before rating. Versioning is explicit so Module 3 can replay or re-rate under historical rules when disputes or audits require it.

Legacy coexistence is staged. During transition, SIMCP can operate as an overlay: legacy clearing for public transport may continue while SIMCP progressively becomes the entitlement and aggregation hub for multimodal caps and cross-mode products. Tolling and parking integrations may start as event mirrors into SIMCP before full presentment shifts.

The integration layer routes each source to the appropriate path (stream versus batch), preserves raw evidence for compliance, and reconciles duplicate or late-arriving events using business keys and deduplication windows agreed per domain.

Partner onboarding is a controlled capability: legal agreement, technical conformance testing against the canonical schema and security profiles, certification in a non-production environment, gradual traffic ramp-up, and operational runbooks for failure modes (back-pressure, replay, manual suspension). Operators receive observability into their own event of acceptance rates and settlement positions without exposing other partners’ commercially sensitive data.

Cross-reference: Detailed non-functional targets for latency, availability, and security controls are specified in Section 6. That section defines the architectural seams and information contracts those targets apply to.

6. Non-Functional Requirements

6.1 Performance & Availability

The SIMCP platform must support real-time mobility operations across multiple transport modes, ensuring fast processing, high reliability, and continuous service availability. Given the large volume of mobility events generated during peak periods, the system must be designed to handle high throughput while maintaining low latency. The key performance and availability requirements are as follows:

- 6.1.1 Throughput:** The system must support high volumes of concurrent mobility events during peak periods without performance degradation.
- 6.1.2 Availability:** Core platform services must achieve a minimum of 99.999% uptime, ensuring uninterrupted access for users and operators.
- 6.1.3 Scalability:** The system must support horizontal scalability to accommodate growth in users and integrated mobility services.
- 6.1.4 Reliability:** The platform must ensure no data loss during event processing and maintain data consistency across all modules.
- 6.1.5 Disaster Recovery:** The system must support defined recovery objectives, including a Recovery Time Objective of ≤ 1 hour and a Recovery Point Objective of ≤ 15 minutes, to ensure timely restoration of services and minimal data loss in the event of system failure.

6.2 Security & Privacy

As SIMCP processes sensitive personal, location, and financial data, strong security and privacy controls are required to protect users and ensure compliance with government regulations. The key security and privacy requirements are as follows:

- 6.2.1 Authentication & Authorisation:** The system must implement secure authentication using OAuth 2.0 / OpenID Connect, with RBAC.
- 6.2.2 Data Encryption:** All data must be encrypted both in transit (TLS) and at rest using secure key management mechanisms.
- 6.2.3 Privacy Protection:** Personal and location data must be handled in accordance with privacy regulations, ensuring data minimisation and restricted access.
- 6.2.4 Audit Logging:** All critical actions must be recorded in secure audit logs to support traceability and dispute resolution.
- 6.2.5 Fraud Detection:** The system must detect abnormal or suspicious activities, such as unusual travel patterns or unauthorised access attempts.

6.3 Usability

The SIMCP platform must support efficient and effective interaction for transport operators to perform operational tasks and manage mobility services. The key operational usability requirements are as follows:

- 6.3.1 Usability Measurement:** The platform achieve an Organizational Net Promoter Score of at least +30 from client side (e.g., TransportNSW).
- 6.3.2 Consistency of Operations:** System processes and workflows must be consistent across all transport services to reduce complexity and errors.

6.4 Maintainability & Operability

The system must be designed to support efficient operation, monitoring, and maintenance, ensuring long-term sustainability and reliability. The key maintainability and operability requirements are as follows:

- 6.4.1 Monitoring:** The system must provide real-time monitoring of performance, errors, and system health.

- 6.4.2 Logging:** Centralised logging must be implemented to support debugging and incident investigation.
- 6.4.3 Fault Handling:** The system must automatically detect and respond to failures.
- 6.4.4 Deployment:** The platform must support CI/CD for rapid updates.

6.5 Future Readiness

The SIMCP platform must be designed to support future expansion and evolving mobility technologies. The key future readiness requirements are as follows:

- 6.5.1 Extensibility:** The system must support integration with new mobility services.
- 6.5.2 Modularity:** A modular architecture must be adopted to allow independent updates of system components.
- 6.5.3 Interoperability:** The platform must support API-based integration with external systems.
- 6.5.4 Configuration Flexibility:** Business rules, pricing policies, and concession criteria must be configurable without requiring system redevelopment, enabling rapid policy updates and regulatory changes.
- 6.5.5 Innovation Support:** The system must be capable of supporting advanced mobility detection capabilities, such as geofencing and route matching, with the ability to improve accuracy over time
- 6.5.6 Data Strategy:** The platform must support scalable data storage and analytics capabilities, enabling long-term data retention, real-time processing, and advanced analytics use cases.

8. Ethical Considerations

The SIMCP platform operates at the intersection of digital identity, financial charging and real-time location tracking across the entire population of NSW. As such, the ethical implications extend beyond technical compliance and *must be addressed as a deliberate design commitment embedded across five Modules*. Below, the section identifies key ethical concerns raised by the proposed platform design, followed by outlines of system-built-in mitigations to address them.

Surveillance and Location Tracking (Module 2)

Phase 3 geofencing introduces passive location monitoring without explicit user action. The platform requires informed consent through Module 1 before any device-based sensing is activated, and raw location traces are not retained beyond the processing window.

Wrongful Charges from Inferred Events (Module 2)

Confidence scoring means some events are based on inference rather than direct confirmation. The platform adopts a resident-first principle, meaning low-confidence events are excluded from billing where evidence is insufficient to justify a charge. However, all denied events are retained in an immutable audit log and monitored by Module 5 for anomalous patterns. Repeated or suspicious low-confidence signals are flagged for investigation, ensuring the dispute pathway cannot be exploited as a mechanism to avoid legitimate charges. Any flagged account is subjected to *human review* before any action is taken. The account holder is also notified in accordance with the platform's transparency obligations. A structured dispute pathway remains available to account holders for any denied event they believe was incorrectly classified.

Pricing Fairness and Equity (Module 3)

Dynamic pricing risks disproportionately burdening users with limited travel flexibility. Non-negotiable price caps, fare floors, and concession protections are enforced at the system level and cannot be overridden by dynamic pricing logic.

Algorithmic Transparency and Accountability (Module 3 and Module 5)

Model-assisted pricing recommendations may not be immediately interpretable to residents. All charges are traceable to a versioned policy rule with plain-language reason codes, and no algorithmic recommendation can alter production charges without human approval.

Data Minimisation and Purpose Limitation (Modules 1, 2, and 5)

The platform aggregates identity, location, and payment data across the population. Data collected by SIMCP is used solely for mobility billing and transport planning. Secondary use requires separate legal authority and is outside the platform's governance scope.

Opt-Out and Accessibility (Modules 1 and 2)

Not all residents own smartphones or consent to location tracking. Tap-based payment remains a fully supported first-class alternative in all phases, and no resident is disadvantaged for choosing not to participate in device-based sensing.

Bias Prevention in Analytics (Module 1 and Module 5)

Pricing feedback loops risk encoding geographic or demographic bias. Module 1 embeds concession categories and eligibility classifications at the identity level, ensuring equity protections are applied consistently from the point of account registration. Module 5 produces distributional impact analyses across defined cohorts, and an independent equity audit is required under Success Criterion 30 as a condition of Phase 3 production readiness.

9. Implementation Approach & Roadmap

9.1 Delivery Phases

We propose a phased delivery approach to keep the program actionable in a government setting, reduce integration risk, and allow benefits to be realised early. The phases are designed to start with a solid foundation, deliver core value in Greater Sydney first, then expand statewide and progressively increase optimisation capability.

Phase 0 - Program Foundation (Set-up & Alignment)

- Goal: Confirm scope, governance, and delivery boundaries before building.
- Key outcomes:
 - Confirm Phase 1 scope (Greater Sydney Area), service inclusions (Opal modes + tolls/regio/selected parking), and agreed payment option(s).
 - Establish governance model (policy change controls, audit requirements, human oversight) and service ownership boundaries (what sits with government vs platform).
 - Define target operating model (roles for agencies, councils, operators, and support/disputes routing).
 - Confirm data sharing principles (minimum required data, access controls, and reporting requirements).

Phase 1 - Foundation Build (Platform + Integration Baseline)

- Goal: Build the platform skeleton and integration baseline that everything else relies on.
- Key outcomes:
 - Identity/account integration approach (Service NSW as the entry point, single account concept).
 - Event ingestion pipelines for core services (public transport usage events, toll transactions, parking events where available).
 - Initial rules framework for pricing/policy configuration (rule versioning, audit trail, approval workflow).
 - Billing view foundation: unified transaction history + monthly statement generation (even if payments remain “as-is” for some services).
 - Security, privacy, logging, and monitoring baselines.

Phase 2 - Core Modes & Billing + Future Capability Build (MVP in Greater Sydney)

- Goal: Deliver a usable MVP for Greater Sydney while starting development of advanced “frictionless travel” capabilities.
- Key outcomes:

- Core mobility coverage: Opal network (train/metro/bus/light rail/ferry) with a digital credential concept (tap-to-travel experience).
- Government/council fees: implement the agreed approach for tolls/rego/selected parking.
- Payment options implemented as agreed:
 - Option A (all partners): per-transaction payments + unified monthly statement
 - Option B (gov/council only): monthly bill via Service NSW payment gateway, with government handling disbursement
- Operational readiness: dispute pathways, incident processes, and support routing and audit logging
- Technology build (develop/proof of concept (PoC)):
 - Geofencing capability (prototype):
 - detect when a user enters a station/stop area (geofence entry)
 - trigger “trip start” signals (event generation)
 - Route matching capability (prototype):
 - track movement patterns along transport corridors (privacy-preserving where possible)
 - identify the most likely line/service used to support accurate trip attribution
 - *Note: In Phase 2, these capabilities are built and tested (e.g., limited pilots, selected lines/zones) but are not yet relied on as the primary validation method.*

Phase 3 - Expansion & Optimisation + Frictionless Travel Rollout (NSW-wide)

- Goal: Expand coverage statewide and implement advanced validation to reduce tap friction where feasible and authorised.
- Key outcomes:
 - Expand from Greater Sydney to statewide NSW (prioritise high-volume corridors and areas with strong data readiness).
 - Add partner services as opt-in integrations (e.g., bike-share, ride-share) starting with booking + receipts/trip history, then deeper integration where feasible.
 - Implement geofencing + route matching:
 - operationalise station-area detection and trip start/end inference
 - deploy route matching models to improve accuracy of trip attribution
 - introduce controls for error handling, fallbacks, and manual correction/disputes
 - Enhanced real-time decisioning (incident-aware responses and mode-balancing recommendations) within governance guardrails.
- End-state vision:
 - No tapping required
 - Reduced waiting time and less friction in boarding
 - Accurate fare calculation supported by reliable trip detection and attribution, with strong governance, privacy protection, and auditability.

9.2 High-Level Project Plan

9.2.1 Milestones

Table 25. Milestones of SIMCP System

Phase 0 — Foundation / Alignment		
M1	Program mobilisation & governance established	Decision structure agreed, roles and responsibilities confirmed and change approval approach defined.
M2	Scope and success measures locked for Phase 1	Phase 1 boundary confirmed (Greater Sydney), in-scope services agreed, and MVP definition documented.

M3	Payment model boundaries confirmed	Option A vs option B agreed in principle, including ownership for billing and disputes.
M4	Data access and integration plan agreed	Access pathways and data-sharing principles confirmed for Opal events, tolling, parking, and incident/network condition feeds.
M5	Target Operating Model drafted	Support model, disputes/refunds routing, and escalation paths defined (what sits with platform vs government vs operators).
Phase 1 — Platform Foundation Build		
M6	Architecture and integration patterns signed off	API/event standards, security controls, logging/monitoring approach, and integration of catalogue baseline approved.
M7	Core event ingestion baseline live	Public transport usage events + toll transactions + initial parking events ingested into the platform in a consistent event schema.
M8	Policy/rules baseline implemented	Versioned policy rules, approval of workflow, audit logging, and rollback capability available.
M9	Transaction history / monthly statement prototype	A working monthly statement view and transaction history aggregation.
M10	Security, privacy, and observability baseline ready	Encryption, RBAC, secure APIs, monitoring, and audit log retention in place for pilot readiness.
Phase 2 — Core Modes & Billing + MVP Pilot (Greater Sydney)		
M11	Core GSA services integrated	Opal ecosystem coverage in place, plus agreed gov/council services.
M12	Payment Option live	Option A: Per-transaction payments with unified statement working end-to-end for applicable services. Option B: monthly bill via Service NSW payment gateway functioning; government disbursement responsibility clearly separated from platform scope.
M13	Operational readiness complete	Dispute handling, refunds routing, incident playbooks, and support processes ready for pilot operations.
M14	Geofencing PoC completed	Station/stop geofence entry detection and “trip start” event triggering tested in limited zones.
M15	Route matching PoC completed	Route inference validated against selected known journeys; accuracy thresholds and exception handling defined.
M16	Pilot 1 completed	Controlled MVP pilot executed, with monitoring and operational feedback captured.
M17	Pilot evaluation and tuning completed	Rules of thresholds refined, exception patterns reviewed, and rollout readiness assessment documented.
M18	Go/No-Go decision for Phase 3	formal decision on production rollout of geofencing/route matching and expansion plan, including required governance controls and fallbacks.
Phase 3 — Expansion & Optimisation + No-tap Rollout (NSW-wide)		
M19	NSW expansion wave 1 delivered	Coverage expanded beyond GSA along prioritised corridors/regions with confirmed data readiness.
M20	Geofencing production rollout	Geofencing enabled in production for supported areas with monitoring, fallbacks (tap remains available), and dispute pathways.
M21	Route matching production rollout	Route matching is deployed to support accurate trip attribution and fare calculation, including error thresholds and manual correction options.
M22	Progressive no-tap rollout achieved	No-tap boarding enabled where feasible with reduced friction, supported by governance controls and service-level monitoring.
M23	Partner add-ons enabled (optional)	Bike-share/ride-share integrated starting with booking + receipts/history; deeper payment integration only if partnerships mature.

9.2.2 Dependencies

Governance & stakeholder alignment

- Cross-agency agreement on scope and responsibilities
Blocks: payment implementation, operating model, pilot readiness.
- Approval of policy guardrails and change controls (human oversight, audit trail requirements, rollback).
Blocks: real-time pricing controls, safe deployment of rule changes.

Data access & quality

- Access to Opal usage events and required metadata
Blocks: end-to-end MVP, accurate statement generation, route matching validation.
- Access to tolling transactions and parking events.
Blocks: gov/council service integration depth, Option B billing scope.
- Incident/network condition feeds (for real-time triggers and operational decisions).
- Blocks: the “real-time” value proposition and optimisation usefulness.

Payments & finance operations

- Service NSW payment gateway integration readiness for monthly billing (Option B).
Blocks: Option B implementation timeline and reconciliation processes.
- Government invoicing and reconciliation capability (monthly bill issuance, collections, exception handling, payout cycles).
Blocks: Option B operational feasibility and rollout confidence.

Technology feasibility (no-tap capabilities)

- User consent and privacy approach for location-based signals (geofencing and route matching require clear governance).
Blocks: PoC execution, production rollout approval.
- Accuracy thresholds and operational fallbacks (tap fallback, dispute/correction workflows).
Blocks: safe implementation of no-tap and public acceptance.

Partner readiness

- Commercial and technical readiness of private providers (Uber/Lime) for booking and data sharing, and later deeper payment integration.
Blocks: Phase 3 partner add-ons

Delivery & change management

- Customer support and dispute handling model across multiple services.
Blocks: pilot go-live, public launch confidence.
- Comms and adoption planning (especially if introducing no-tap behaviours).
Blocks: user trust and operational stability.

9.2.3 Gantt Chart

Table 26. Gantt Chart for SIMCP Project Timeline

Phase	Month	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Foundation/Alignment	1. Program mobilisation & governance setup																								
	2. Confirm scope + success metrics																								
	3. Confirm payment model boundaries																								
	4. Data access planning & agreements																								
	5. Operating model design																								
Foundation Platform Build	6. Solution architecture + integration patterns																								
	7. Event ingestion baseline																								
	8. Rules & policy engine baseline																								
	9. Billing view foundation																								
	10. Security/privacy baseline + logging/monitoring																								
	11. Test environment + data quality checks																								
Core Modes & Billing + MVP Pilot	12. Core service integration																								
	13. Payment implementation																								
	14. Operational readiness build																								
	15. Geofencing PoC (build + limited testing)																								
	16. Route matching PoC (build + validation)																								
	17. Pilot 1 (controlled MVP pilot in selected zones/corridors)																								
	18. Pilot evaluation + policy/rules tuning																								
	19. Go/No-Go decision for Phase 3 rollout																								
	20. Scale rollout: expand coverage beyond GSA																								
Expansion & No-tap Rollout	21. Implement geofencing in production																								
	22. Implement route matching in production																								
	23. No-tap rollout																								
	24. Partner add-ons																								
	25. Ongoing optimisation + reporting uplift																								

10. Risk Analysis & Mitigation

SIMCP risk management assumes that technical, organisational, and political failure modes interact: a poorly communicated price change can amplify distrust even when systems are stable, while brittle integrations can force manual workarounds that themselves create security and fairness risks. The register below lists each risk with impact, likelihood, mitigation, and a named owner. Severity scales are qualitative (High / Med / Low) for proposal-stage planning. In addition to the controls embedded in each module, SIMCP treats risk management as a continuous governance process across delivery phases. The register is reviewed at phase gates (Section 9) and used to prioritise controls that protect residents and operational stability first: privacy and consent readiness before any location-sensing expansion, evidence quality and dispute pathways before any tap-free charging reliance, and strong separation-of-duties before any policy changes can affect charges at scale. Where risks relate to availability, latency, or security posture, mitigations are designed to align with the non-functional targets and controls in Section 6 so that operational measures (SLOs, logging, incident response, and cost guardrails) can be verified, audited, and improved over time.

Full details are provided in the risk register in **Appendix 5 (Table 27)**.

11. Assumptions & Constraints

11.1 Legislative & Privacy Assumptions

The SIMCP program assumes that the NSW Privacy and Personal Information Protection Act will continue to apply during the project. The PIA approved in Phase 0 is expected to remain valid for Phases 1 and 2. However, a new PIA will be needed before Phase 3, especially for location-based features.

It is assumed that SIMCP is allowed to receive identity and vehicle data from Service NSW to create and manage Mobility Accounts. If there are any changes in privacy laws or rules that affect how data is collected, stored, or shared, this will be treated as a risk and must be reviewed before continuing.

11.2 Scope Boundaries

SIMCP focuses on mobility charging and account management. The following are not included in this project:

- 11.2.1 Replace existing billing systems used by operators
- 11.2.2 Managing transport infrastructure or operations
- 11.2.3 Building a full mobile app beyond account and billing features

Payments will be handled by external systems. SIMCP will only generate billing and settlement information.

11.3 Legacy System Constraints

SIMCP integrates with, rather than replaces, existing transport operator systems. The following constraints apply across all delivery phases:

- 11.3.1 **Opal System** - SIMCP consumes Opal tap event data as a read-only feed and must not alter Opal's existing fare calculation or settlement logic. Opal clearing may continue to run in parallel during Phases 1 and 2.
- 11.3.2 **Tolling Infrastructure** - Toll gantry systems operate under private concession agreements and use proprietary formats. Integration will rely on agreed APIs or batch extracts per operator. Existing corporate fleet toll accounts remain outside SIMCP's billing scope in Phases 1 and 2.
- 11.3.3 **Parking Systems** - Technical maturity varies across councils. Only councils able to provide structured event data within agreed formats will be included in the billing scope; others will be onboarded once their systems meet the minimum data standard.
- 11.3.4 **Service NSW Identity Platform**. Module 1 depends on Service NSW's OAuth 2.0/OpenID Connect APIs for identity verification and vehicle data. Any changes to these API contracts during the program will be assessed for impact before affected phases proceed.
- 11.3.5 **External Mobility** - These providers operate proprietary platforms and cannot be compelled to share data. Phase 3 integration is contingent on voluntary commercial partnership agreements and remains out of scope until those agreements are in place.

11.4 Rollout & Timeline Constraints

The project will be delivered in phases, and each phase must be completed and approved before moving to the next. Phase 3 features such as geofencing and route matching depend on the success of Phase 2 testing. A Go/No-Go decision will be made before full rollout.

The timeline may be affected by approval processes between different agencies. Phase 2 testing will only happen in Greater Sydney, while expansion to regional NSW will happen in Phase 3.

The program also assumes that proper communication and user support will be done before each release to help users adopt the system smoothly.

11.5 Cloud Platform Assumptions

Assumes AWS remains the approved cloud platform under NSW Government policy throughout all phases; any change in government cloud strategy would require architecture review.

11.6 Concession Policy Stability

Assumes Transport for NSW concession categories remain stable in structure through Phase 2; any new concession types added mid-program will require a Module 1 and Module 3 change request.

12. Expected Benefits

SIMCP is distinguished by three principles: it **preserves existing infrastructure rather than replacing it**, it **delivers value progressively** so the approach is validated before full statewide commitment, and it is **built citizen-first** with equity, transparency, and opt-in sensing embedded from day one. The following sections translate these principles into concrete benefits for government, residents, and the broader community.

12.1 Benefits to Government

SIMCP gives government the ability to act on its network in real time, something the current fragmented ecosystem makes impossible. Consider a peak-hour incident on the M7, congestion builds, train platforms fill, and the network starts to strain. Today, government has no unified lever to respond. Under the SIMCP operation, the same event triggers an automatic congestion pricing signal in Module 3, a real-time bus fare discount to shift demand, and a live dashboard update in Module 5 and all within minutes, all from a single platform. Settlement across every participating operator runs in one structured cycle at the end of the month rather than dozens of separate arrangements. Dynamic pricing and consolidated settlement give STA **better visibility over revenue flows** across all modes, enabling **more accurate forecasting** and **reducing revenue leakage** from fragmented operator-direct arrangements. And because each capability is proven on live infrastructure before the next phase is committed, government **sees these outcomes from a working system**, not a business case.

12.2 Benefits to Residents

SIMCP transforms a fragmented daily transport experience into a **single, seamless interaction, without asking residents to change how they travel**. Consider a Western Sydney commuter who drives to Parramatta station, parks, takes the train into the city, and walks home from a different station that evening. Today that journey touches three operators, three accounts, and three bills. Under SIMCP it is one itemised entry on one monthly statement, against one identity, with concession entitlements applied automatically. Three shifts make this possible. One account, one bill allowing every charge to be traceable to the originating mobility event for transparent dispute resolution. **Concessions that follow the resident**, not the mode, meaning that entitlements apply automatically across every supported service with no repeated registration. Furthermore, optional and advanced detection is opt-in, tap-based payment remains available, and SIMCP works with existing behaviour rather than demanding new behaviour to deliver value.

12.3 Broader Societal Benefits

SIMCP creates the foundation for a transport system that is more sustainable, and more equitable at the network level. Consider a transport planner trying to understand whether last month's congestion pricing trial on the M2 shifted demand to public transport. Today that answer requires reconciling data from multiple operators across weeks. Under SIMCP, the same analysis is available in Module 5 the following morning with cross-modal, longitudinal, and grounded in **actual travel behaviour rather than operator extracts**. Beyond planning, the platform creates **natural incentives** for more sustainable travel choices without legislation or mandate. Moreover, because **equity protections** and **independent oversight** are embedded directly into the delivery conditions, the benefits of a smarter network are not concentrated among those who already have the most choices.

Taken together, the mentioned benefits are not independent outcomes, they are mutually reinforcing. A government that can oversee and respond to its network in real time makes more equitable decisions for the residents. Residents who experience a fairer, simpler and more transparent systems are more likely to trust and engage with public transport. And a society where pricing signals, equity protections, and data driven planning function as a whole produces a transport network that serves everyone, not just those with the most flexibility. SIMCP is not simply a platform upgrade but also the infrastructure that makes evidence-based equitable and responsive mobility governance possible for NSW.

13. Conclusion

The SIMCP provides a practical and future-ready pathway for transforming NSW's fragmented transport charging ecosystem into a unified, resident-centric mobility platform. Its core value lies in connecting identity, mobility event capture, dynamic pricing, billing, settlement, and analytics into one coordinated system. This enables residents to manage mobility through one trusted account and one clear billing view, while giving government improved visibility, control, and policy agility across the transport network.

The proposed solution is closely aligned with the State Transport Authority's strategic objectives. By enabling real-time pricing, consolidated billing, concession-aware charging, and system-wide analytics, SIMCP supports congestion management, mode shift, equity, administrative simplification, and environmental sustainability. It also maintains compatibility with existing systems through a hybrid payment approach: Option A delivers early value through per-transaction payments and unified monthly statements, while Option B provides a pathway toward deeper monthly billing integration through the Service NSW payment gateway for eligible services.

SIMCP is ready for staged implementation because it avoids a high-risk "big bang" replacement model. The phased roadmap allows the platform to begin with foundational identity, event ingestion, pricing rules, and billing capabilities before progressing to a Greater Sydney MVP and eventual statewide expansion. Future capabilities such as geofencing, route matching, shared mobility integration, and analytics-driven optimisation can be introduced progressively once governance, privacy, and operational controls are proven.

Overall, SIMCP offers a scalable and governable platform that can deliver immediate improvements in transparency and coordination while building toward a more seamless, intelligent, and inclusive mobility ecosystem for NSW. Its clear scope boundaries, phased delivery model, and strong governance controls position the proposal as both strategically aligned and practically achievable.

Appendices

Appendix 1.

This appendix describes how document updates are communicated to STA during the proposal development and review cycle. It is intended to support transparency and traceability for reviewers, rather than define internal editing workflows.

Version history as the source of truth

The Document Control table at the front of this proposal is the authoritative record of changes. Each entry summarises what changed, when, and who led the update.

What changes are highlighted to reviewers

For each new version issued to STA, the team will provide:

- A concise summary of substantive changes (sections/modules impacted)
- Any scope-impacting changes (inclusions/exclusions, assumptions, dependencies)
- Any material changes to risks or mitigations (Section 10)

Configuration management intent

Where SIMCP policies, rules, or reporting definitions change in future delivery (e.g., pricing guardrails, KPI definitions), those changes will follow controlled change governance with versioning, approvals, and auditability consistent with the governance approach described across Modules 3 and 5.

Appendix 2. Terminology and Definitions

A. Canonical phase definitions (baseline: Section 9.1)

Use **exactly these names** in cross-references (success criteria, modules, roadmap).

Phase	Official name (short)	Goal (one line)	Geography / when
0	Program Foundation (<i>Set-up & Alignment</i>)	Lock scope, governance, TOM, payment boundaries, data-sharing principles before major technical build.	Pre-build; no MVP geography
1	Foundation Build (<i>Platform + Integration Baseline</i>)	Platform skeleton: identity path, core ingestion, rules baseline, statement/history prototype, security/observability baselines.	Prepares Greater Sydney as first integration target
2	Core Modes & Billing + Future Capability Build (<i>MVP, Greater Sydney</i>)	Operable MVP (Opal-class + agreed gov/council services, payment Option A/B); PoC geofencing & route matching (not sole validation yet).	Greater Sydney Area
3	Expansion & Optimisation + Frictionless Travel Rollout (<i>NSW-wide</i>)	Statewide expansion; operational geofencing + route matching; progressive no-tap where approved; partner add-ons as partnerships allow.	NSW

Cross-author rule: If a module mentions “Phase 2” or “Phase 3,” add a parenthetical once, e.g. “Phase 2 (Greater Sydney MVP, Section 9.1).”

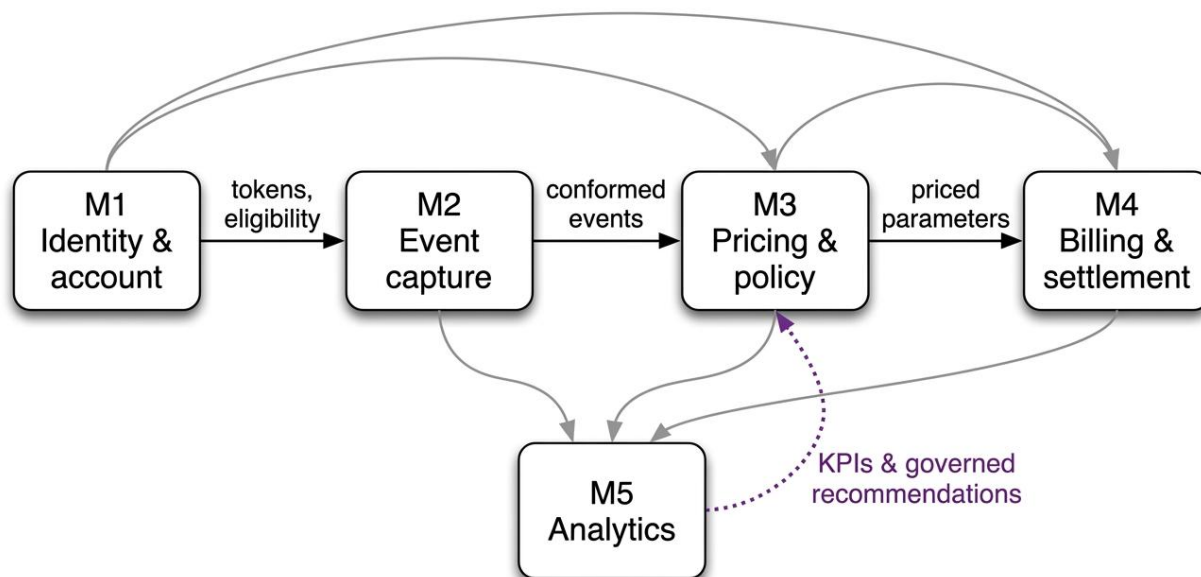
B. Team terminology (quick glossary)

Term	Meaning	Common misuse
Mobility ID	Stable identifier for a person/org context in SIMCP (key for joins).	Calling the whole profile “the ID.”
Mobility Account	Full account record (profile, links, vehicles, devices, roles, concessions).	Interchanging with “Mobility ID.”
Module 2 (official)	Mobility Event Capture & Sensing Engine	Legacy phrase “Usage Detection & Event Capture Engine.”
Linkt	Consumer toll account/product brand; Transurban is the operator group—pick one sentence in Background + stick to it elsewhere or say “Transurban (Linkt).”	Mixing without one clarifying line.
MyE-Toll	Use MyE-Toll everywhere (NSW-style spelling); avoid MyEToll / MyeToll .	Inconsistent spelling in M1.

Option A / Option B	A: per-tx and/or unified statement across partners. B: monthly bill via Service NSW for in-scope gov/council paths + agreed disbursement.	Implying both execute payments inside SIMCP core.
Confidence score	M2 signal reliability metadata (tap > inferred, etc.).	Treating as a marketing term without M4/M5 linkage.

Appendix 3. SIMCP overall workflow (logical data path)

Operational charging path is **M1 → M2 → M3 → M4**; **M5** consumes outputs for measurement and **feeds governed insight** back to policy/pricing owners (not silent production price changes).



SIMCP logical data path (Alconic)

How to read this: Operators still **send** raw feeds; **Module 2** is the place those signals become **one canonical mobility-event stream** for downstream rating and billing. Module 3 “inputs” in prose describe **events from M2** plus **identity/eligibility from M1**, not a parallel universe where M3 only reads operators directly.

Appendix 4. Module 2 Appendix

A. Event Detection and Validation Fall-back Plan and Architecture

If uncertain → manage, degrade, escalate, or defer

Table 1. Fallback Coverage by Failure Type (Module-Level View)

Failure Type	Fallback Strategy	Affected Module 2
Missing signals	Infer using last known state	Module 2 (Event Detection & Construction), Module 3 (Pricing), Module 5 (Analytics)
Low confidence detection	Downgrade / exclude	Module 2, Module 3, Module 4 (Billing), Module 5
Identity resolution failure	Reject or queue for retry	Module 1 (Identity), Module 2
Data ingestion failure	Retry / buffer input streams	Module 2
Processing failure	Revert to rule-based logic	Module 2, Module 3
Publishing failure	Queue events for delayed delivery	Module 2, Module 3, Module 4
Fraud/anomaly detection	Flag for investigation and restrict downstream use	All

Table 2. Fallback Plan for Unsuccessful Identity Binding

Scenario	Trigger Condition	Immediate Action	Fallback Behaviour	Resolution Path	Affected Modules
Device not in Authorised Device Registry	Signal received from unregistered device	Reject signal immediately	Event not constructed or published	User prompted to register device via Module 1 before re-attempting	M1, M2
Mobility ID not resolved	Signal received but cannot be matched to a valid Mobility ID	Queue event for retry within defined time window	If unresolved after retry window, event is rejected and logged to audit trail	Account holder notified to verify account status via Module 1	M1, M2
Vehicle plate not matched to registered account	Toll or parking signal received but plate not found in registered vehicle records	Hold event pending manual lookup	If unresolved, event flagged for operator review — no charge applied	Operator or STA administrator reviews and manually links vehicle record if valid	M1, M2, M4
Account suspended or deactivated	Mobility ID resolved but account status is inactive	Reject signal immediately	Event not constructed or published — no charge applied	Account holder directed to Module 1 to resolve account status before services resume	M1, M2
Identity provider unavailable	Service NSW authentication service unreachable	Queue incoming signals for retry	Fall back to last confirmed identity state for active sessions only	Resume normal identity resolution when Service NSW connectivity is restored	M1, M2

Table 3. Fall-back plan for low confident events

Confidence Band	Trigger Condition	Immediate Action	Fallback Behaviour	Resolution Path	Affected Modules
Medium confidence	Score falls below high threshold but above denial threshold	Hold event await corroborating signal within defined time window (e.g. 30–60 seconds)	If corroborating signal arrives → re-score and accept; if not → apply conservative fallback pricing rule based on minimum defensible fare	Event flagged in audit log with reason code; account holder may raise dispute via Module 4 if charge is contested	M2, M3, M4
Low confidence	Score falls below denial threshold	Deny event immediately and exclude from billing pipeline	Event retained in immutable audit log with confidence score, detection method, and reason code	Account holder notified of denied event; structured dispute pathway available via Module 4; Module 5 monitors for repeated denial patterns on same account	M2, M4, M5
Repeated low confidence on same account	Module 5 detects pattern of repeated low-confidence signals from same Mobility ID or device	Flag account for investigation	Downstream use of future events from flagged account restricted pending human review	Human reviewer assesses whether pattern indicates genuine detection failure or potential fraud; account holder notified in accordance with transparency obligations	M2, M4, M5
All inferred events — system-wide confidence degradation	Module 5 detects platform-wide drop in confidence scores across multiple corridors or modes	Alert operations team	Revert to conservative rule-based classification across affected zones; tap fallback enforced where available	Operations team investigates root cause (e.g. infrastructure outage, GPS interference); normal confidence scoring resumes once issue is resolved	M2, M3, M5

B. Confidence Scoring Mechanism

When the system detects a mobility event, it doesn't just ask *what happened*, it also asks *how sure we are that this actually happened?* That certainty level is the confidence score.

Rather than a simple hierarchy, think of confidence as a score out of 100 built from several components. Here's a starting framework to react to:

Table 1

Factor	Max points	Logic
Signal source type	40	Tap=40, GPS+route match=25, Geofence only=15, Speed pattern only=10
Signal corroboration	30	How many independent signals agree? 3+=30, 2=20, 1=0 A GPS trace corroborated by a geofence trigger is like two credible witnesses, strong but not perfect
Spatial plausibility	15	Is the location reachable from last known position in the time elapsed?
Temporal consistency	15	Does timing make sense given prior events in this session?

Exemplars:

- A tap-on with matching GPS and plausible location = $40+20+15+15 = 95$
- A geofence only trigger with no corroboration and borderline timing = $15+0+10+5 = 30$

What this score actually means:

The score falls into three bands, and each band triggers a different system response:

Band	What it means	What the system does
High (75-100)	Strong evidence	Charge automatically, pass to M3
Medium (40-74)	Uncertain	Wait for more signals, apply conservative fallback if unresolved
Low (0-39)	Insufficient evidence	Deny the event, do not charge

C. Phase 3 Event Detection and Sensing Engine - Implementation of Geo-fencing and Route Matching:

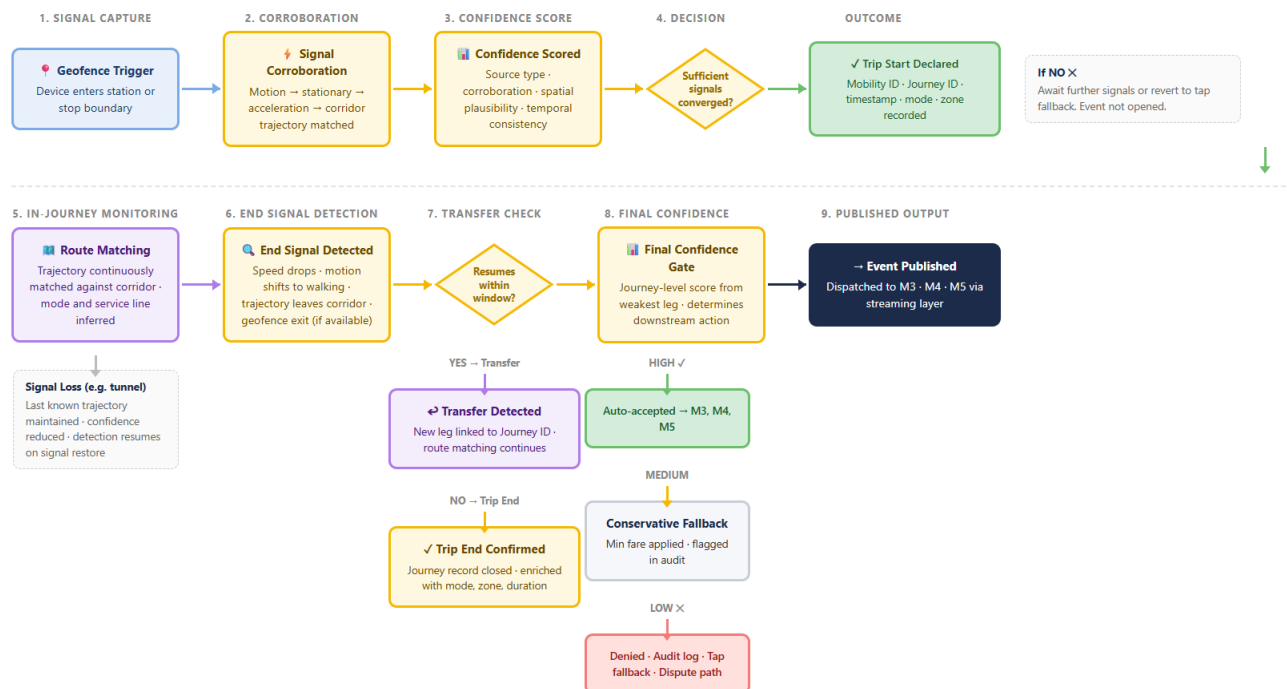
In Phase 3, the platform introduces geofencing-based trip detection at selected stations on a testing basis. Unlike tap-based detection where trip boundaries are explicitly defined by a user action, geofencing-based detection infers trip boundaries from a convergence of passive signals rather than explicit user actions. The following principles govern how the system determines when a journey starts and ends.

Trip Start. A trip start is declared when multiple independent signals agree within a short time window, typically a geofence boundary trigger, a motion pattern shift from walking to stationary, a subsequent acceleration to transport-consistent speed, and a trajectory matching a known corridor. A geofence trigger alone is insufficient; the system requires corroboration before committing to a trip start.

Trip End. A trip end is inferred from a reduction in speed, a shift to walking motion, and divergence from the transport corridor. Where a destination geofence is available, it serves as the primary anchor. Where it is not, corridor departure and speed reduction are used, resulting in a lower confidence score for that event.

Transfers. Following a candidate trip end, the system applies a dynamic transfer window check rather than a fixed time threshold. The transfer window is informed by real-time network conditions including scheduled service frequencies, known delays, trackwork disruptions, and the transport mode involved. For example, a metro transfer may close within five minutes while a regional train connection may warrant a significantly longer window. Where service disruption data is available from network feeds, the window is extended accordingly. If the user resumes transport-consistent movement within the applicable window, the event is classified as a transfer and linked to the existing journey record rather than closing it.

Fallback. Tap-on and tap-off remains available in all Phase 3 scenarios. Where geofencing detection produces a low-confidence outcome, the system reverts to tap-based confirmation.



Appendix 5. Section 10 Appendix

Table 27. SMICP Risk Analysis and Mitigation

ID	Category	Risk	Impact	Likelihood	Mitigation (summary)	Owner
R1	Security	Cyber breach of mobility data	High	Med	Zero-trust segmentation; strong IAM/PAM; encrypt in transit/at rest; central logging + SIEM; breach playbooks + notifiable breach alignment; pen test/red team; minimise data in analytics exports.	STA CISO / SIMCP Security Lead
R2	Security	Insider threat / privileged misuse	High	Low	PAM for admin roles; JIT access; dual control on sensitive exports; immutable audit logs; periodic access recertification; targeted monitoring on high-risk actions.	STA CISO / SIMCP Security Lead
R3	Security	Key management / encryption misconfiguration	High	Low	Central KMS controls; standardised key policies; automated compliance checks; rotate keys/secrets; encryption-by-default guardrails; break-glass procedures.	STA CISO / SIMCP Security Lead
R4	Security	DDoS or volumetric attack impacts availability	High	Med	Edge rate limiting/WAF; auto-scaling; isolate partner blast radius; degrade gracefully; incident runbooks and regular game days.	SIMCP Platform Owner / SRE Lead
R5	Privacy & Legal	Privacy / legislative change invalidates Phase 3 location-sensing assumptions	High	Med	Refresh DPIA/PIA before expansion; configurable minimisation modes (retention/granularity); maintain compliant tap-only contingency; regulator engagement; clear lawful basis documentation.	STA Legal–Privacy / SIMCP Executive Sponsor
R6	Privacy & Legal	Re-identification risk in transparency reporting	High	Low	Small-cell suppression; minimum cohort thresholds; geo/time generalisation; review process for any external publication; independent privacy review for new dashboards/exports.	STA Legal–Privacy / Data Governance Lead
R7	Privacy & Legal	Consent UX failure / poor comprehension for location-based signals	Med	Med	Plain-language consent; staged opt-in pilots; “what data is used” explainers; easy opt-out where lawful; measure consent drop-off and adjust.	Customer Experience Lead / STA Legal–Privacy

R8	Integration	Service NSW / identity integration failure or data exposure (Module 1)	High	Med	Integration SLAs; staged rollout; synthetic monitoring; break-glass recovery; least-privilege scopes; secrets rotation; DPIA updates for scope changes.	STA CISO / SIMCP Security Lead + Module 1 Service Owner
R9	Integration	Integration failure with operators / partners	High	Med	Conformance labs + certification; phased onboarding with volume ramp; contractual SLAs/credits; dual-run; replayable event stores; cutover war-room.	SIMCP Integration Program Manager
R10	Integration	Schema/version drift breaks downstream modules	High	Med	Explicit schema versioning; contract tests; compatibility matrix; deprecation windows; automated validation gates; rollback plans.	SIMCP Integration Program Manager
R11	Data Quality	Duplicate/late events cause incorrect charging or reporting	High	Med	Dedupe windows; idempotency keys; late-arrival handling; reconciliation jobs; alerting; replay capability with audit trails.	SIMCP Platform Owner / Data Engineering Lead
R12	Data Quality	Reference data errors (zones, topology, concessions) distort outcomes	High	Med	Master-data stewardship; change control; validation rules; shadow updates; monitored KPI anomalies; rapid rollback of reference-data releases.	STA Data Governance Lead / Module Owners
R13	Analytics	KPI definition inconsistency across teams undermines trust	Med	Med	Versioned “gold” logic; KPI catalogue; lineage documentation; governance approvals for KPI changes; periodic stakeholder sign-off.	Module 5 Lead / Data Governance Lead
R14	Model / Optimisation	Model or optimisation bias harms protected cohorts	High	Low	Model risk management; bias testing; documented assumptions; monitoring for disparate impact; independent review; human accountability for any charge-impacting decisions.	STA Legal–Privacy / Model Risk Owner
R15	Operational	Tap-on → tap-free technology transition risk (Module 2 Phase 2→3)	High	Med	Conservative confidence thresholds; shadow-mode before charging; consent UX; corridor benchmarking; clear fallback to tap when low confidence.	SIMCP Platform Owner / STA Legal–Privacy + Module 2 Lead
R16	Operational	Low-confidence tap-free capture drives disputes/unfair charging (SC27 linkage)	High	Med	If corridor accuracy/confidence below threshold (e.g., SC27 92%), force fallback to tap-based eligibility; fast-track dispute/refund workflow;	SIMCP Platform Owner / Customer Operations Lead + Module 2 Lead

					publish confidence/dispute metrics.	
R17	Finance	Third-party payment / PSP interface failure or payment-data leakage (Module 4)	High	Med	PCI-aligned delegation; tokenisation (no PAN); webhook signing; reconciliation controls; vendor security attestations; chaos testing for partial settlement.	SIMCP Platform Owner / Finance Operations Lead
R18	Finance	Settlement/reconciliation errors impact operator trust	High	Med	Multi-stage reconciliation; tolerances + exception queues; audit-ready reason codes; independent sampling audits; dispute runbooks with operators.	Finance Operations Lead / Module 4 Lead
R19	Governance	Governance/approval latency blocks time-sensitive pricing changes	High	Med	Decision SLAs + escalation; pre-approved guardrails for delegated changes; approval queue dashboard; stale recommendation expiry; re-baselining.	STA Policy Board secretariat / SIMCP Platform Owner
R20	Governance	Separation-of-duties failure (policy authors deploy unreviewed changes)	High	Low	SoD controls; signed policy bundles; immutable audit logs; independent review for material changes; canary releases + rollback.	STA Policy Board secretariat / SIMCP Platform Owner
R21	Change & Adoption	Public resistance / trust failure	High	Med	Co-design; plain-language explainability; opt-in pilots; transparent KPI reporting; accessible dispute channels; independent oversight of equity metrics.	STA Communications / Customer Experience Lead
R22	Change & Adoption	Political backlash	High	Med	Early briefing packs; documented equity impact assessments; staged rollout; “no surprises” governance; scenario planning for inquiries.	STA Executive Sponsor / Government Relations
R23	Delivery	Phased rollout / dual-run incompatibility (Section 9 phases)	High	Med	Strict interface versioning; feature flags; dual-run exit criteria; test harness; legacy vs SIMCP data reconciliation; migration runbooks.	SIMCP Integration Program Manager
R24	Operability	Observability gaps slow incident response	Med	Med	End-to-end tracing; SLOs; central logs/metrics; on-call runbooks; regular incident simulations; clear ownership boundaries.	SRE Lead / SIMCP Platform Owner

R25	Performance	Latency or throughput misses break “real-time” value proposition	High	Med	Capacity testing; back-pressure; partitioning by operator/geography; load shedding; prioritise core modes; measure P95/P99 and tune.	SIMCP Platform Owner / SRE Lead
R26	Cost	Analytics/streaming cost blowout under peak volumes	Med	Med	Cost guardrails; tiered retention; lifecycle policies; sampling for non-critical telemetry; optimise query patterns; budget alerts with governance.	SIMCP Platform Owner / Finance Operations Lead